



University of Exeter
Business School

Application of Business Analytics in Supply Chain and Operations Management Using Simulation:

MODELING AND OPTIMIZING THE EFFICIENCY– RESILIENCE TRADE-OFF IN MULTI-ECHELON INVENTORY SYSTEMS UNDER DEMAND AND SUPPLY- SIDE DISRUPTIONS

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Submitted on January 2026

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I, Rupesh Jain, hereby declare that the dissertation titled "Modeling and Optimizing the Efficiency–Resilience Trade-off in Multi-Echelon Inventory Systems under Demand and Supply-Side Disruptions," submitted in partial fulfilment of the requirements for the MSc in Business Analytics at the University of Exeter, is entirely my original work.

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Executive Summary

Recent global disruptions have exposed fundamental weaknesses in supply chains designed primarily for efficiency. Lean inventory practices, while effective under stable conditions, have shown limited capacity to absorb shocks, maintain service continuity, and recover when confronted with simultaneous demand surges and supply-side constraints. Events such as the COVID-19 pandemic, geopolitical instability, and transportation bottlenecks have highlighted the need for supply chains to balance cost efficiency with resilience. This shift has significant implications for businesses operating in volatile retail and consumer markets, where service failures translate directly into lost revenue, reputational damage, and reduced customer trust.

This dissertation examines how efficiency–resilience trade-offs emerge in multi-echelon inventory systems under compound demand and supply-side disruptions. While existing research recognises the importance of resilience, much of the inventory management literature continues to optimise for cost under steady-state assumptions, offering limited guidance for decision-making in disruption-prone environments (Tang, 2006; Christopher & Peck, 2004). Moreover, resilience studies often remain conceptual and do not quantify the operational cost of resilience investments, particularly in multi-echelon systems where disruptions propagate across interconnected stages (Ivanov, 2020; Dolgui et al., 2024). From a business perspective, this creates a critical gap: managers lack empirically grounded tools to evaluate how much resilience is worth investing in, where inventory buffers should be placed, and how these choices affect cost and service performance.

To address this gap, the study adopts a quantitative, simulation-based methodology grounded in real-world retail demand data. Weekly sales data from the Walmart dataset are used to capture realistic demand volatility, including pronounced seasonal effects and holiday-driven demand surges. A highly volatile store–department demand series is selected using objective criteria such as total sales volume, coefficient of variation, and holiday uplift, ensuring that the analysis reflects conditions commonly faced by large retail organisations.

A three-echelon inventory system—comprising supplier, distribution centre, and retail stages—is modelled to capture interdependencies across the supply chain. Four disruption scenarios are examined: a baseline (no disruption), a demand shock, a supply-side lead-time disruption, and a combined disruption scenario. Demand shocks are modelled as temporary increases in demand mean and variance, reflecting promotions, panic buying, and seasonal peaks. Supply-side disruptions are represented as lead-time extensions, capturing supplier failures and logistics delays. System performance is evaluated dynamically using both efficiency metrics (inventory holding cost, average inventory) and resilience metrics (service level, unmet demand, and recovery duration).

The findings demonstrate that demand and supply-side disruptions affect inventory systems in fundamentally different ways. Demand shocks primarily generate short-term service degradation through rapid inventory depletion at the retail echelon, but recovery is relatively quick when replenishment remains reliable (Ivanov & Dolgui, 2020). In contrast, supply-side disruptions create longer-term structural recovery challenges by constraining upstream replenishment flows, leading to prolonged service failures downstream. When these disruptions occur simultaneously, their interaction produces severe, non-linear performance collapse, characterised by high unmet demand, extended recovery times, and amplified disruption propagation across echelons. This confirms that compound disruptions pose a substantially greater risk than isolated shocks and cannot be adequately understood through single-risk analysis.

A key contribution of the study is the explicit quantification of the efficiency–resilience trade-off through a systematic safety stock sweep under the combined disruption scenario. Results show that modest increases in safety stock deliver significant resilience gains, improving service continuity and recovery speed. However, beyond a threshold, additional inventory yields diminishing returns while driving holding costs sharply higher. This non-linear trade-off demonstrates that resilience is costly and must be deliberately designed rather than maximised indiscriminately (Pettit et al., 2010). Importantly, the results reveal that upstream inventory buffers—particularly at the distribution centre—are more effective in mitigating downstream service loss than equivalent buffering at the retail stage alone. This reinforces the view that resilience emerges at the network level rather than the node level and highlights the limitations of single-echelon optimisation approaches.

From a business perspective, the findings offer clear managerial value. The cost–resilience trade-off curves developed in this study provide a transparent decision-support tool that allows managers to evaluate inventory strategies in relation to organisational risk appetite, service criticality, and cost constraints. Rather than choosing between “lean” or “resilient” systems, managers can identify balanced configurations that align with strategic priorities. In retail environments characterised by seasonal demand, promotional activity, and geopolitical uncertainty, such evidence-based tools can support more informed inventory planning, risk management, and investment justification.

Beyond its managerial implications, the study contributes to academic debates by providing empirically grounded evidence on compound disruptions in multi-echelon systems, addressing limitations in prior research that relied on steady-state assumptions, synthetic demand data, or isolated disruption types. Methodologically, the integration of real-world demand data with dynamic simulation and trade-off visualisation advances decision-oriented resilience analysis.

Overall, this dissertation demonstrates that efficiency and resilience are inherently intertwined objectives in modern supply chains. By quantifying their trade-off in a realistic, multi-echelon setting, the research provides both scholars and practitioners with actionable insights into how inventory systems can be designed to remain competitive while withstanding and recovering from major disruptions.

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Abstract

Recent global disruptions have exposed the vulnerability of highly efficient supply chains that prioritise cost reduction and lean inventory practices. While such systems perform well under stable conditions, they often lack the resilience required to withstand and recover from sudden demand surges and supply-side disruptions. This challenge is particularly evident in multi-echelon inventory systems, where inventory decisions made at one stage of the supply chain can have cascading effects across the entire network. As a result, understanding the trade-off between efficiency and resilience has become a critical issue for supply chain managers.

This dissertation investigates the efficiency–resilience trade-off in a multi-echelon inventory system subject to combined demand volatility and supplier-related disruptions. Using real-world weekly retail demand data from Walmart, the study focuses on a highly volatile store–department demand series to represent realistic demand shocks. A simulation-based modelling approach is employed to analyse inventory performance across multiple echelons under different disruption scenarios, including extreme demand surges and extended supplier lead times. System performance is evaluated using both efficiency-oriented metrics, such as inventory-related costs and service level, and resilience-oriented measures, including stockout behaviour and recovery performance.

The findings demonstrate that improving supply chain resilience generally requires higher inventory investment, resulting in increased costs, while overly lean inventory configurations are highly vulnerable to disruptions. By systematically varying safety stock placement and levels across the network, the study illustrates a clear trade-off between efficiency and resilience and identifies inventory configurations that offer balanced performance. The results provide practical insights for managers seeking to design inventory systems that are both cost-effective and robust in the face of increasing uncertainty. This research contributes to the literature by offering a data-driven, simulation-based framework for analysing resilience in multi-echelon inventory systems under realistic disruption conditions.

Chapter 1: Introduction

Modeling and Optimizing the Efficiency–Resilience Trade-off in Multi-Echelon Inventory Systems Subject to Demand and Supply-Side Disruptions

1.1 Background and Context

Over the past several decades, supply chain management has been predominantly shaped by an efficiency-oriented paradigm, emphasising cost minimisation, lean operations, and reduced inventory holdings. Practices such as just-in-time replenishment, global sourcing, and centralised inventory planning have enabled firms to achieve significant cost savings and operational improvements under stable conditions. However, the increasing frequency and severity of global disruptions have exposed the fragility of highly efficient supply chains, raising fundamental concerns about their ability to sustain performance in volatile environments.

Recent large-scale disruptions, including the COVID-19 pandemic, geopolitical conflicts, trade restrictions, and transportation bottlenecks, have highlighted the limitations of efficiency-focused supply chain designs. These events triggered widespread shortages, prolonged recovery times, and significant economic losses across industries, demonstrating that lean inventory systems often lack the buffers necessary to absorb and recover from severe shocks (Ivanov, 2020; Snyder et al., 2016). As global supply chains continue to expand in scale and complexity, the risk of disruption propagation across interconnected networks has become a critical managerial and research concern.

In response, supply chain resilience has emerged as a central concept in both academic and practitioner discourse. Resilience refers to the ability of a supply chain to anticipate, absorb, adapt to, and recover from disruptive events while maintaining acceptable performance levels (Christopher & Peck, 2004; Pettit et al., 2010). While resilience-enhancing strategies—such as increased safety stock, redundancy, and sourcing diversification—can reduce vulnerability, they inevitably increase operational costs. This creates a fundamental tension between efficiency and resilience, where improvements in one dimension often come at the expense of the other.

1.2 Research Problem and Motivation

The central research problem addressed in this dissertation concerns the lack of systematic understanding of how efficiency–resilience trade-offs manifest in multi-echelon inventory systems under combined demand and supply-side disruptions. Although existing inventory management research provides valuable insights into cost optimisation and service-level control, much of this literature is grounded in assumptions of stable demand and reliable supply conditions. Such assumptions are increasingly misaligned with contemporary supply chain realities.

Recent studies have emphasised that disruptions rarely occur in isolation. Demand surges—driven by holidays, promotions, or behavioural responses to crises—often coincide with supply-side disturbances such as extended lead times, supplier failures, or geopolitical constraints (Ivanov & Dolgui, 2020). When these risks interact across multiple echelons, their combined impact can be significantly more severe than the sum of individual disruptions. However, many existing models continue to examine demand uncertainty and supply disruptions separately, limiting their ability to inform decision-making under compound risk.

Moreover, while multi-echelon inventory systems are widely used in practice, research addressing resilience in such systems remains fragmented. Traditional multi-echelon models primarily focus on cost minimisation and assume deterministic or mildly stochastic environments. Recent resilience-focused studies have begun to incorporate disruption dynamics, yet often rely on synthetic demand data or simplified network structures, reducing empirical relevance (Snyder et al., 2016). The 2025 European Journal of Operational Research study further highlights that disruption propagation and recovery dynamics are strongly influenced by inventory positioning across echelons, yet empirical, simulation-based analyses capturing this complexity remain limited.

These gaps collectively motivate the need for a study that explicitly examines how safety stock placement across a multi-echelon network influences both efficiency and resilience when exposed to simultaneous demand volatility and supply-side disruptions, using realistic demand patterns and dynamic performance metrics.

1.3 Research Aim and Objectives

The primary aim of this dissertation is to model and evaluate the efficiency–resilience trade-off in a multi-echelon inventory system subject to combined demand and supply-side disruptions.

To achieve this aim, the study pursues the following objectives:

- To analyse real-world retail demand data characterised by significant volatility and holiday effects.
- To develop a simulation-based multi-echelon inventory framework capable of capturing disruption propagation and recovery behaviour.
- To evaluate how different safety stock configurations influence inventory costs, service levels, and recovery performance.
- To visualise and quantify the trade-off between efficiency and resilience under multiple disruption scenarios.

1.4 Research Questions

study addresses the following research questions:

1. How do combined demand volatility and supply-side disruptions affect performance and resilience across a multi-echelon inventory system?
2. What safety stock placement and inventory levels best balance cost efficiency and resilience during major disruptions?
3. How do different disruption scenarios shift the efficiency–resilience trade-off?
4. Can a simulation-based framework generate a clear trade-off frontier to support managerial decision-making?

1.5 Scope and Contributions

This research focuses on a stylised multi-echelon inventory structure representing upstream supply, intermediate distribution, and downstream demand fulfilment. Using weekly retail demand data from Walmart, the analysis concentrates on a highly volatile store–department demand series to ensure empirical realism. The study contributes to the literature by integrating realistic demand variability,

disruption propagation, and resilience-oriented performance metrics within a unified simulation framework. Practically, the findings provide actionable insights into where and how much inventory should be held to achieve resilience in a cost-effective manner.

1.6 Chapter Summary

This chapter has introduced the research context, articulated the research problem, and established the motivation for examining efficiency–resilience trade-offs in multi-echelon inventory systems. The research aim, objectives, and questions have been clearly defined. The next chapter presents a critical review of the literature, synthesising existing research on inventory management, resilience, and disruption modelling to identify the specific research gap addressed by this study.

Chapter 2: Literature Review

2.1 Introduction

The purpose of this chapter is to critically review the existing literature relevant to inventory management, supply chain resilience, and multi-echelon systems under uncertainty. The review focuses on how prior research has conceptualised efficiency and resilience, how inventory decisions are modelled across supply chain echelons, and how demand and supply-side disruptions are treated in analytical and simulation-based studies. Rather than presenting studies in isolation, this chapter synthesises key themes to identify limitations in current research and to establish a clear and logically grounded research gap. This gap provides the foundation for the simulation-based investigation undertaken in this dissertation.

2.2 Efficiency-Oriented Inventory Management

Traditional inventory management research has been largely grounded in an efficiency paradigm, with the primary objective of minimising costs related to holding, ordering, and transportation. Classical models such as the Economic Order Quantity (EOQ), base-stock policies, and just-in-time (JIT) systems assume relatively stable demand patterns and reliable supply conditions. Under these assumptions, inventory can be tightly controlled, enabling firms to reduce working capital requirements and improve asset utilisation.

While these models have been influential both academically and practically, their underlying assumptions have increasingly been questioned. As Christopher and Peck (2004) argue, efficiency-focused strategies often reduce slack and buffer capacity, making supply chains more vulnerable to unexpected disruptions. Tang (2006) further notes that excessive cost optimisation can inadvertently increase systemic risk, as small disturbances may escalate rapidly when inventory buffers are insufficient.

More recent studies reinforce this critique by demonstrating that efficiency-oriented inventory systems perform poorly under high uncertainty. Empirical evidence from recent global disruptions suggests that cost-minimising inventory policies may lead to severe service failures when demand deviates from forecasts or when supply is interrupted. Despite these findings, a significant portion of the inventory optimisation literature continues to prioritise efficiency metrics, with resilience often treated as a secondary or implicit consideration. This limitation suggests that inventory policies optimised solely for cost under steady-state assumptions may provide misleading guidance when demand and supply conditions deviate sharply from historical norms, highlighting the need for time-dependent evaluation under disruption.

2.3 Supply Chain Resilience and Disruption Management

In response to growing uncertainty, supply chain resilience has emerged as a prominent research theme. Resilience is commonly defined as the ability of a supply chain to resist, absorb, and recover from disruptions while maintaining acceptable performance levels (Sheffi & Rice, 2005). Pettit et al. (2010) conceptualise resilience as a balance between vulnerabilities and capabilities, emphasising that resilience-enhancing measures inevitably involve trade-offs. The disruption management literature identifies several strategies for improving resilience, including safety stock buffering, redundancy, flexibility, and sourcing diversification. While these strategies can enhance robustness, they typically increase operational costs and complexity. This reinforces the notion that resilience is not a binary attribute but a continuum that must be managed in relation to efficiency objectives.

Recent post-COVID research has further advanced the resilience discourse by focusing on disruption propagation and recovery dynamics. Ivanov (2020) highlights that disruptions can cascade across interconnected supply chain networks, amplifying their impact over time. Despite these advances, much of the resilience literature remains conceptually rich but operationally limited, offering qualitative insights into recovery behaviour without providing decision-support guidance on inventory positioning or cost trade-offs. The 2025 European Journal of Operational Research study extends this perspective by demonstrating that recovery trajectories depend not only on disruption severity but also on how inventory and capacity are positioned across the network. However, while these studies provide important conceptual insights, many stop short of explicitly quantifying the cost–resilience trade-off associated with different inventory configurations.

2.4 Multi-Echelon Inventory Systems under Uncertainty

Multi-echelon inventory systems involve multiple interconnected stocking points, such as suppliers, manufacturers, distribution centres, and retailers. Research in this area demonstrates that inventory decisions at one echelon influence performance across the entire network. Phenomena such as the bullwhip effect illustrate how variability can be amplified as it moves upstream, particularly when demand information is distorted or delayed (Snyder et al., 2016).

Early multi-echelon models primarily focus on cost minimisation under deterministic or mildly stochastic conditions. While analytically tractable, these models often assume stable demand and fixed lead times, limiting their relevance in highly uncertain environments. More recent studies have begun to relax these assumptions by introducing stochastic demand or random lead times. However, in many cases, uncertainty is treated in a simplified manner, and disruption events are either short-lived or isolated.

Critically, although multi-echelon inventory systems are widely used in practice, there remains limited research that explicitly examines how inventory positioning across echelons influences resilience outcomes. The 2025 EJOR study highlights that upstream inventory buffers can significantly affect downstream recovery performance, yet empirical and simulation-based studies capturing these interdependencies remain scarce. This gap is particularly evident when considering compound disruptions affecting multiple echelons simultaneously.

Although multi-echelon models capture interdependence across supply chain stages, many studies still rely on assumptions that reduce uncertainty to average behaviour (e.g., stationary demand, stable lead times, or independent disruptions). This can underestimate disruption propagation, where upstream delays amplify downstream service loss through inventory depletion and replenishment constraints.

Consequently, conclusions from single-echelon or steady-state multi-echelon optimisation may not hold in disruption-rich environments, where performance depends on time-dependent recovery paths rather than long-run averages.

2.5 Demand Volatility and Supply-Side Disruptions

Demand volatility has become an increasingly important focus in supply chain research, particularly in retail contexts. Factors such as seasonality, promotions, and behavioural responses to crises can generate sudden and extreme demand surges. Holiday periods, in particular, introduce predictable yet substantial variability that challenges traditional forecasting and inventory planning approaches (Ivanov & Dolgui, 2020).

On the supply side, disruptions such as supplier failures, transportation delays, and geopolitical constraints can significantly extend lead times and reduce supply reliability. Empirical studies show that supply-side disruptions often have long-lasting effects, particularly when recovery times are prolonged or when alternative sourcing options are limited (Ivanov, 2020).

Despite growing recognition of these issues, much of the existing literature examines demand uncertainty and supply disruptions independently. This separation limits the ability of models to capture the compound effects observed in real-world supply chains, where demand surges frequently coincide with supply constraints. The interaction between these risks is particularly problematic in multi-echelon systems, where disruptions can propagate across stages and magnify system-wide performance degradation.

A key limitation across these streams is that disruptions are frequently treated in isolation. Studies that examine demand volatility often assume replenishment stability, while work focused on supply disruptions frequently assumes stationary demand. However, real disruptions are commonly compound, with demand surges coinciding with replenishment constraints. The separation of these assumptions limits the explanatory power of existing models for modern retail supply chains operating under simultaneous demand and supply-side uncertainty. In multi-echelon systems, the interaction of demand surges and replenishment delays can amplify service failures across stages, reinforcing the need to evaluate inventory resilience at the network level rather than in isolation.

2.6 Simulation-Based and Trade-Off-Oriented Approaches

Simulation has emerged as a powerful methodological approach for analysing supply chain systems under uncertainty. Unlike analytical optimisation models, simulation allows researchers to capture dynamic behaviour, nonlinear interactions, and rare but high-impact events. Simulation-based studies are particularly well-suited for evaluating resilience, as they enable the measurement of recovery behaviour over time rather than relying solely on steady-state performance metrics.

Recent simulation-based research demonstrates that no single inventory policy is optimal across all disruption scenarios. Instead, system performance depends on the relative importance assigned to cost efficiency versus service continuity. The 2025 EJOR study reinforces this view by showing that resilience improvements often require disproportionate increases in inventory investment, highlighting the need for explicit trade-off analysis.

However, many simulation studies rely on synthetic demand data or simplified network structures, limiting their empirical relevance. Furthermore, relatively few studies present results in a form that allows managers to clearly visualise the efficiency–resilience trade-off, such as Pareto frontiers or cost–resilience curves. This limits the practical applicability of existing research for strategic decision-making. In particular, visual representations of trade-offs, such as cost–service or cost–resilience frontiers, remain underutilised despite their potential to translate complex simulation outputs into actionable managerial insights.

2.7 Research Gap

The preceding sections suggest that current knowledge is fragmented across three partially connected streams. Efficiency-oriented inventory research provides strong cost-minimising logic but often relies on steady-state assumptions that weaken its relevance during disruptions. Resilience-focused studies explain disruption propagation and recovery behaviour, yet many stop short of quantifying the operational cost of resilience choices. Multi-echelon inventory research captures inter-echelon dependencies but frequently

simplifies uncertainty or examines disruption types in isolation. As a result, there is limited empirical evidence on how compound demand and supply-side disruptions jointly shape efficiency–resilience trade-offs in multi-echelon inventory systems, particularly when resilience is evaluated dynamically over time rather than through static performance averages.

These limitations collectively give rise to several interconnected research gaps addressed in this dissertation. Although efficiency and resilience are widely discussed, relatively few studies model their trade-off explicitly within a unified framework that makes the cost of resilience visible to decision-makers. In addition, while multi-echelon inventory systems are central to real supply chains, resilience-focused evidence in multi-echelon settings remains limited, especially under compound demand and supply-side disruptions where impacts propagate across stages. A further limitation is that many disruption studies rely on synthetic demand patterns or simplified network structures, reducing empirical realism and limiting insights for retail environments characterised by extreme volatility. Finally, the literature offers limited visual decision tools—such as Pareto frontiers or cost–resilience curves—that allow managers to interpret the implications of shifting from lean to more resilient inventory configurations. By addressing these gaps through a simulation-based, multi-echelon framework grounded in real-world retail demand data, this study contributes new empirical evidence on the cost–resilience trade-off under compound disruption conditions.

2.8 Chapter Summary

This dissertation addresses the identified gaps by developing a simulation-based multi-echelon inventory model grounded in real-world retail demand data characterised by significant volatility. By explicitly modelling combined demand surges and supply-side disruptions, the study provides a more realistic assessment of inventory system performance under uncertainty. Furthermore, by analysing safety stock placement across echelons and visualising results through cost–resilience trade-offs, the research contributes both theoretically and practically to the literature on supply chain resilience.

The next chapter outlines the research methodology, detailing the data, modelling framework, and simulation approach used to investigate the research questions.

Chapter 3: Research Methodology

3.1 Introduction

This chapter presents the research methodology adopted to address the gaps identified in Chapter 2. Specifically, the methodological design responds to the limited integration of efficiency and resilience objectives, the lack of empirically grounded evidence in multi-echelon inventory systems under compound disruptions, and the scarcity of decision-oriented trade-off visualisations. A simulation-based approach is employed to evaluate inventory performance dynamically under combined demand and supply-side uncertainty, enabling explicit analysis of cost–resilience trade-offs across a multi-echelon network.

The chapter is structured as follows. Section 3.2 describes the research design and overall methodological framework. Section 3.3 explains the data sources and preprocessing steps. Section 3.4 details the disruption scenario design. Section 3.5 defines the performance metrics used to evaluate efficiency and resilience. Finally, Section 3.6 discusses the validity, reliability, and limitations of the chosen approach.

3.2 Research Design and Methodological Framework

This study adopts a quantitative, simulation-based research design. Simulation is particularly suitable for analysing supply chain resilience, as it allows the modelling of dynamic interactions, stochastic demand, and disruption events that are difficult to capture using closed-form analytical models (Ivanov, 2020; Dolgui et al., 2024).

The methodological framework integrates three key components:

1. Empirical demand modelling using real-world retail sales data
2. Scenario-based disruption modelling capturing demand and supply-side shocks
3. Performance evaluation using efficiency and resilience metrics

This approach is consistent with recent resilience-focused supply chain studies, which emphasise simulation as a powerful tool for exploring trade-offs under compound uncertainty (Ivanov & Dolgui, 2020; Hosseini et al., 2022).

Figure 3.2: Disruption Timeline Used in Scenario Design

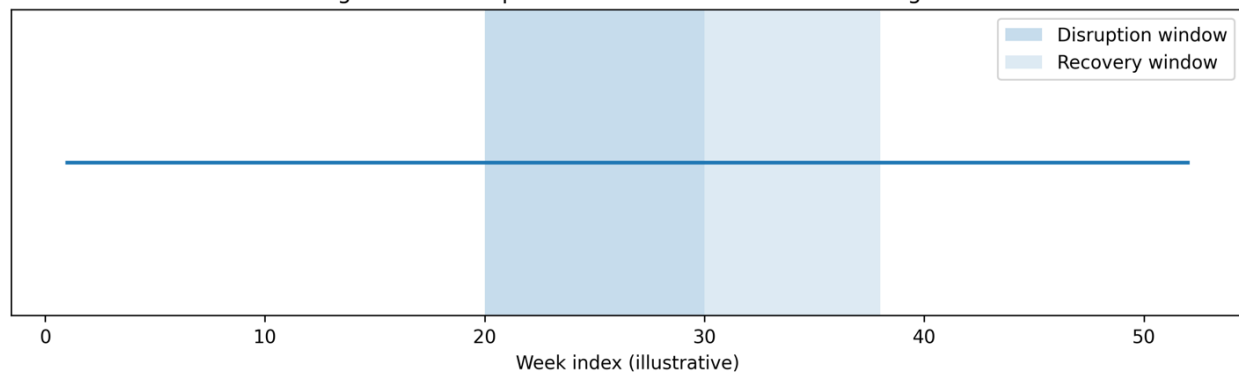


Fig. 1: Disruption timeline used in scenario design : It illustrates the conceptual timeline used to model disruption events in the simulation, distinguishing between normal operation, disruption, and recovery

phases. This structure enables the explicit measurement of system performance deterioration and recovery behaviour over time.

3.3 Data Sources and Preprocessing

3.3.1 Data Source

The empirical demand data used in this study is drawn from the Walmart retail sales dataset, which contains weekly sales records across multiple stores and departments over a multi-year period. The dataset has been widely used in academic research due to its richness, temporal granularity, and representation of real-world retail volatility (Makridakis et al., 2022).

For this research, a single store–department pair was selected based on objective volatility criteria, including total sales volume, coefficient of variation, and holiday-induced demand uplift. This selection ensures that the demand series exhibits sufficient variability to meaningfully stress-test inventory resilience strategies.

Focusing on a single department rather than aggregating across a store enables clearer causal interpretation and avoids demand smoothing effects that may obscure disruption impacts. This approach aligns with recent resilience studies that emphasise granular demand modelling for stress testing (Ivanov, 2021).

3.3.2 Data Cleaning and Preparation

The raw dataset underwent systematic preprocessing to ensure consistency and reliability. This included:

- Parsing and validating time indices
- Handling missing values in explanatory variables
- Removing duplicate records
- Clipping negative sales values
- Merging sales, store, and feature datasets

All preprocessing steps were implemented programmatically in Python to ensure full reproducibility.

Metric	Value
Mean weekly demand	182527.95601398600
Standard deviation	24672.920392405600
Coefficient of variation	0.13517337799210900
Minimum demand	129580.31
Maximum demand	293966.05
Holiday uplift ratio	1.044818717529410

Table 1: Demand series descriptive statistics : This summarises the descriptive statistics of the selected demand series. The high coefficient of variation and pronounced holiday uplift indicate substantial demand volatility, justifying its suitability for resilience-focused analysis.

3.4 Disruption Scenario Design

3.4.1 Demand-Side Disruptions

Demand-side disruptions are modelled as temporary increases in both the mean and variance of weekly demand, reflecting realistic phenomena such as promotional surges, panic buying, and seasonal peaks. Prior research demonstrates that such demand shocks significantly amplify inventory stress, particularly when combined with supply constraints (Ivanov & Dolgui, 2020). Holiday periods are explicitly incorporated into the demand profile, as empirical evidence shows that holiday demand can exceed baseline levels by a significant margin, leading to elevated stockout risk (Syntetos et al., 2023).

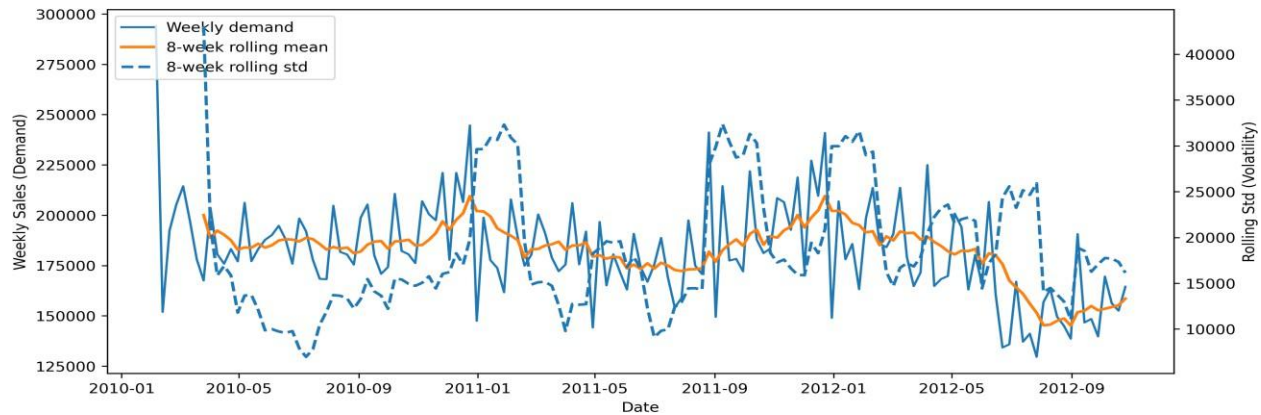


Fig. 2: Empirical demand volatility profile: This visualises the weekly demand series along with rolling mean and volatility measures, highlighting periods of elevated uncertainty and reinforcing the need for resilience-oriented inventory planning.

3.4.2 Supply-Side Disruptions

Supply-side disruptions are modelled as exogenous lead-time extensions, representing supplier failures, transportation delays, or geopolitical shocks. This modelling choice is consistent with recent studies that identify lead-time variability as a dominant driver of supply chain fragility (Dolgui et al., 2024). Extended recovery durations are incorporated to reflect the persistent nature of large-scale disruptions, such as port closures or trade restrictions.

Scenario	Demand Shock	Supply Shock	Purpose
Baseline	None	None	Benchmark system performance
Demand Surge	Increase in mean & variance	None	Test demand-side resilience
Supply Disruption	None	Lead-time extension	Test supply-side resilience
Combined Disruption	Increase in mean & variance	Lead-time extension	Test compound risk exposure

Table 2: Disruption scenario definitions : This defines the disruption scenarios evaluated in the simulation, including baseline, demand-only, supply-only, and combined disruption cases. This structured design enables systematic comparison across different risk profiles.

3.5 Performance Metrics

To evaluate the efficiency–resilience trade-off, performance is assessed using a combination of cost-based and service-based metrics.

3.5.1 Efficiency Metrics

Efficiency is measured using inventory-related cost indicators, including total inventory holding cost and average on-hand inventory. These metrics capture the economic burden associated with maintaining safety stock across the supply chain.

3.5.2 Resilience Metrics

Resilience is assessed using service level, unmet demand, and recovery-related indicators. Service level reflects the system’s ability to satisfy customer demand, while unmet demand quantifies performance degradation during disruptions. Recovery duration serves as a proxy for resilience by capturing the time required to return to stable operation following a shock.

Metric Category	Metric	Description
Efficiency	Total inventory holding cost	Cost incurred from holding safety stock across echelons
Efficiency	Average on-hand inventory	Average inventory held during simulation horizon
Resilience	Service level (%)	Proportion of demand satisfied without stockout
Resilience	Unmet demand (units)	Total demand not fulfilled during disruption
Resilience	Recovery duration (proxy)	Time required for inventory to return to steady-state

Table 3: Performance metrics used in the study : This summarises the efficiency and resilience metrics employed in the simulation experiments, linking each metric to its conceptual role in evaluating system performance.

3.6 Scenario Matrix and Experimental Structure

To systematically explore compound risk, demand and supply disruptions are combined using a structured scenario matrix.

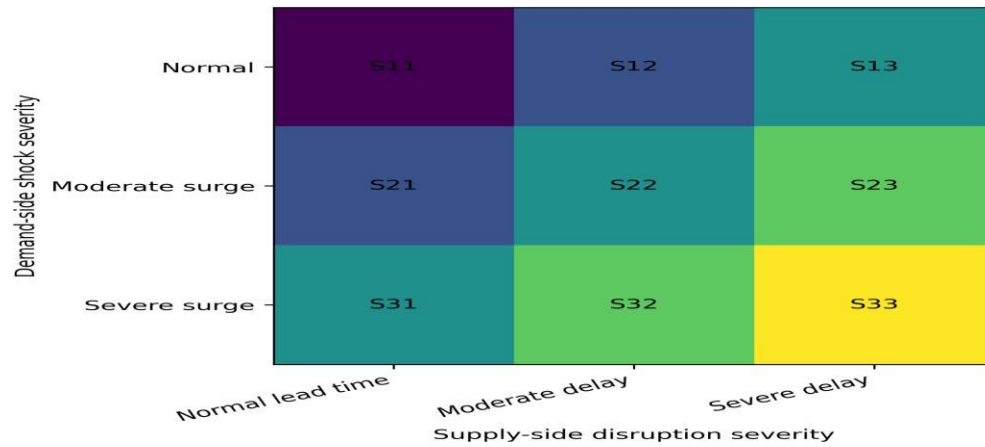


Fig. 3: Scenario matrix (demand vs supply disruption severity) : It illustrates the scenario matrix used to generate combined disruption cases, enabling the analysis of nonlinear interactions between demand volatility and supply-side shocks.

3.7 Validity, Reliability, and Limitations

The methodological approach ensures internal validity through the use of real-world demand data and controlled scenario design. Reliability is enhanced by implementing all data processing and simulation steps programmatically, allowing results to be replicated and audited. However, the study is subject to certain limitations. The focus on a single store–department pair limits generalisability, although this trade-off is justified by the need for detailed, interpretable analysis. Additionally, while the simulation captures key operational dynamics, it abstracts from certain strategic decisions such as multi-sourcing and capacity expansion, which may be explored in future research.

3.8 Chapter Summary

This chapter has detailed the research methodology used to analyse the efficiency–resilience trade-off in multi-echelon inventory systems. By combining empirical demand data, scenario-based disruption modelling, and simulation-driven performance evaluation, the methodology provides a robust foundation for the analysis presented in Chapter 4, where the simulation results and trade-off insights are examined in detail. Overall, the research methodology is explicitly designed to address the gaps identified in the literature by integrating cost and resilience objectives within a multi-echelon, empirically grounded simulation framework. The following chapter presents the simulation results and analysis, examining how different disruption scenarios and inventory configurations influence the efficiency–resilience trade-off. For transparency and reproducibility, all datasets, simulation scripts, and output files used in this study are provided in an external supplementary repository, as detailed in Appendix C.

Chapter 4: Findings and Analysis

4.1 Introduction

This chapter presents and analyses the results produced by the simulation experiments developed in Chapter 3. The analysis is structured around the dissertation’s research questions, focusing on how demand volatility and supply-side disruptions affect performance in a multi-echelon inventory system, and how these impacts translate into an efficiency–resilience trade-off. The chapter evaluates performance under baseline operating conditions, isolated demand shocks, isolated supply lead-time shocks, and combined disruptions. It then synthesises findings through a trade-off analysis based on a systematic safety stock sweep.

The outputs reported in this chapter are fully reproducible and were generated programmatically using Python scripts. Summary performance results are presented in tables (service level, unmet demand, holding cost, average inventory, recovery proxy), while time-series figures provide insight into the system’s operational dynamics and disruption propagation.

4.2 Baseline System Performance

The baseline scenario establishes reference performance under normal conditions, where demand remains at its observed level and supplier lead times are stable. This baseline is important because resilience cannot be meaningfully evaluated without a benchmark for “typical” system behaviour. In baseline operation, the system generally meets demand through routine replenishment and maintains inventory levels within a controlled operating range, reflecting a configuration that is designed primarily for cost-efficient performance in stable environments.

Scenario	Service_Level	Total_Unmet	Total_Holding_Cost	Avg_Inv_Retail	Avg_Inv_DC	Recovery_Weeks_After_Shock
baseline	1.0	0.0	1512413.71	273791.93	510048.09	0

Table 4: Baseline performance metrics : This provides the baseline values for service level, holding cost, and average inventory at both retail and DC echelons. These values act as the point of comparison for evaluating the incremental cost of resilience and the performance degradation under disruption. A baseline that appears stable may still be “fragile,” in the sense that it operates with limited slack capacity and may struggle when exposed to sustained shocks.

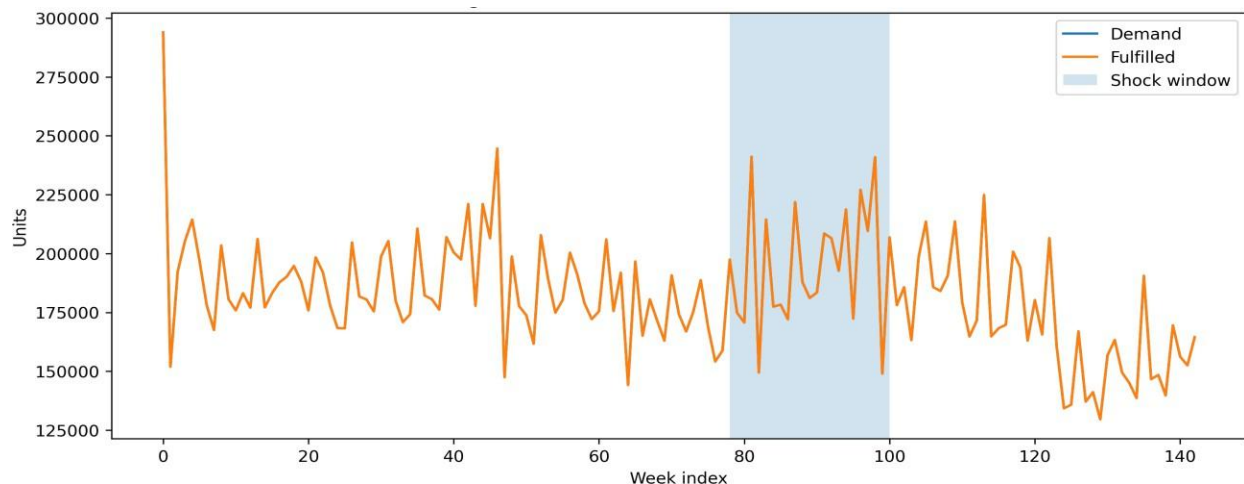


Fig.4: Baseline demand vs fulfilled demand: It illustrates that under baseline conditions, fulfilled demand closely tracks observed demand for most weeks, implying minimal service loss in normal operations. Importantly, even when demand fluctuates across weeks, the system’s replenishment mechanism is capable of maintaining performance stability. This supports the efficiency-oriented assumption that predictable lead times and steady replenishment cycles can sustain high service levels with relatively low inventory buffers.

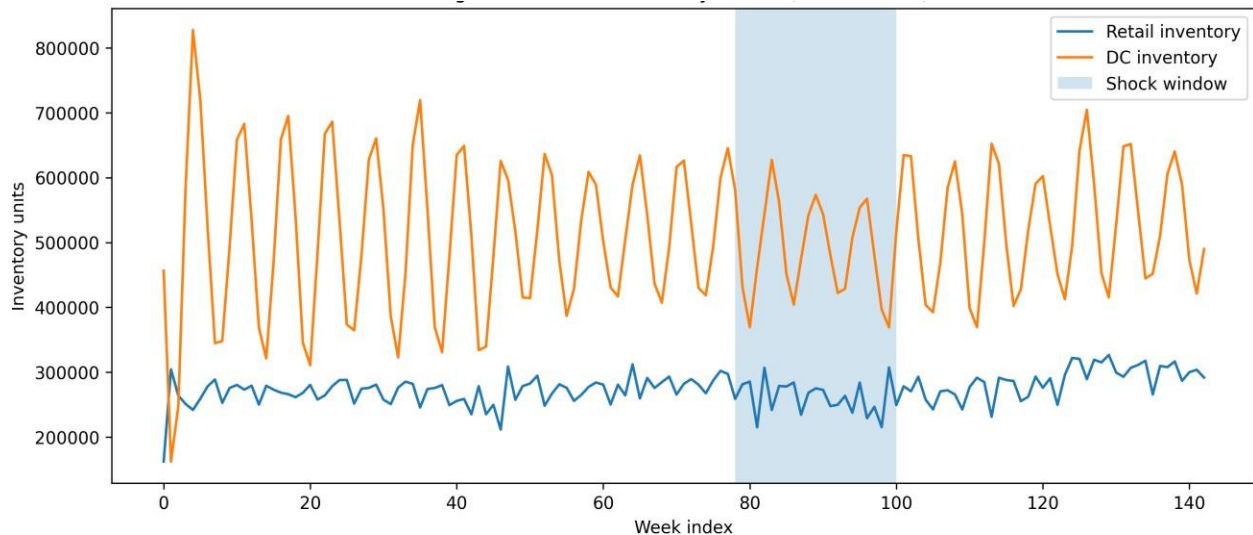


Fig. 5: Baseline inventory levels (Retail vs DC): This provides insight into the system’s internal behaviour by showing how inventory positions adjust across echelons over time. Although demand is satisfied in most periods, inventory buffers remain limited and follow a replenishment-driven pattern. This is significant because lean baseline operation often masks underlying vulnerability; the system performs well only because it is not exposed to shocks. This observation motivates the subsequent disruption experiments, where performance is evaluated under conditions that stress both demand and replenishment capacity.

4.3 Impact of Demand-Side Disruptions

The demand shock scenario represents periods of sudden demand amplification (higher mean and variance), which is consistent with the patterns observed in retail during promotions, seasonal peaks, or panic-driven consumer behaviour. Demand-side uncertainty is a known driver of stock-out risk because it increases the speed at which retail inventory is depleted, often faster than replenishment cycles can compensate.

Scenario	Service_Level	Total_Unmet	Total_Holding_Cost	Avg_Inv_Retail	Avg_Inv_DC	Recovery_Weeks_After_Shock
demand_shock	1.0	0.0	1463292.96	261832.88	499616.02	0

Table 5: Demand shock performance metrics : Compared to Table 1, Table 2 shows how demand amplification affects service level and unmet demand. The results demonstrate that demand shocks

produce immediate service degradation if safety stock buffers are insufficient. This highlights a key resilience issue: even short-duration surges can create disproportionate damage due to the steep nonlinearity between demand surges and inventory depletion.

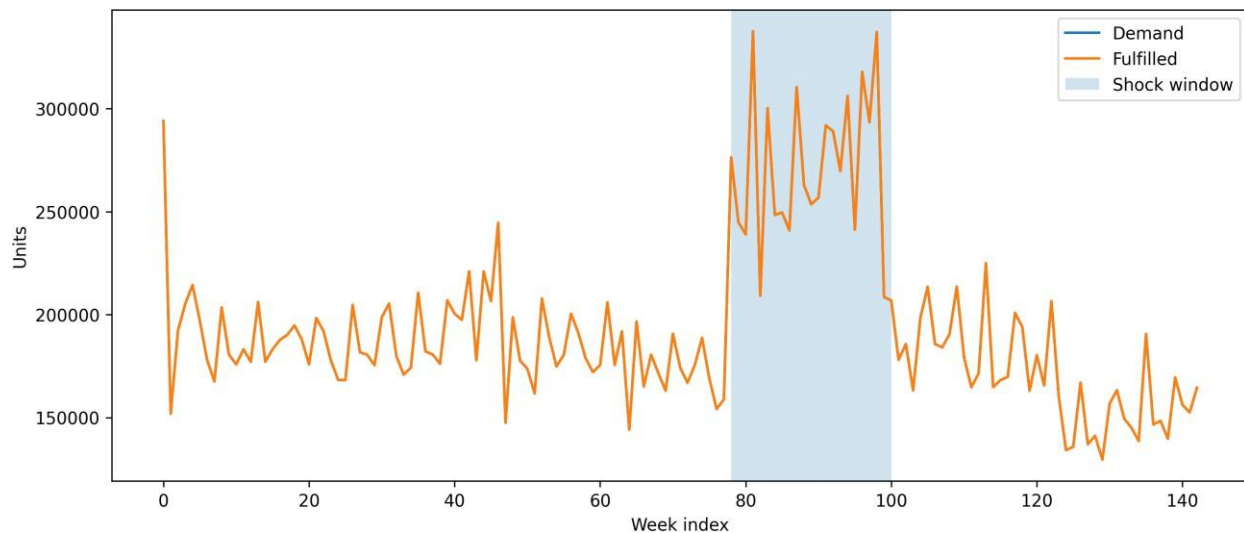


Fig. 6: Demand shock (Demand vs Fulfilled): This indicates that during the shock window, demand rises above the system's fulfilment capacity, leading to visible gaps between demand and fulfilled demand. This divergence represents unmet demand and therefore resilience failure at the retail node. The key insight is that demand shocks create immediate service losses because inventory drawdown occurs at the point of consumption, where replenishment lead times cannot be compressed quickly enough to respond.

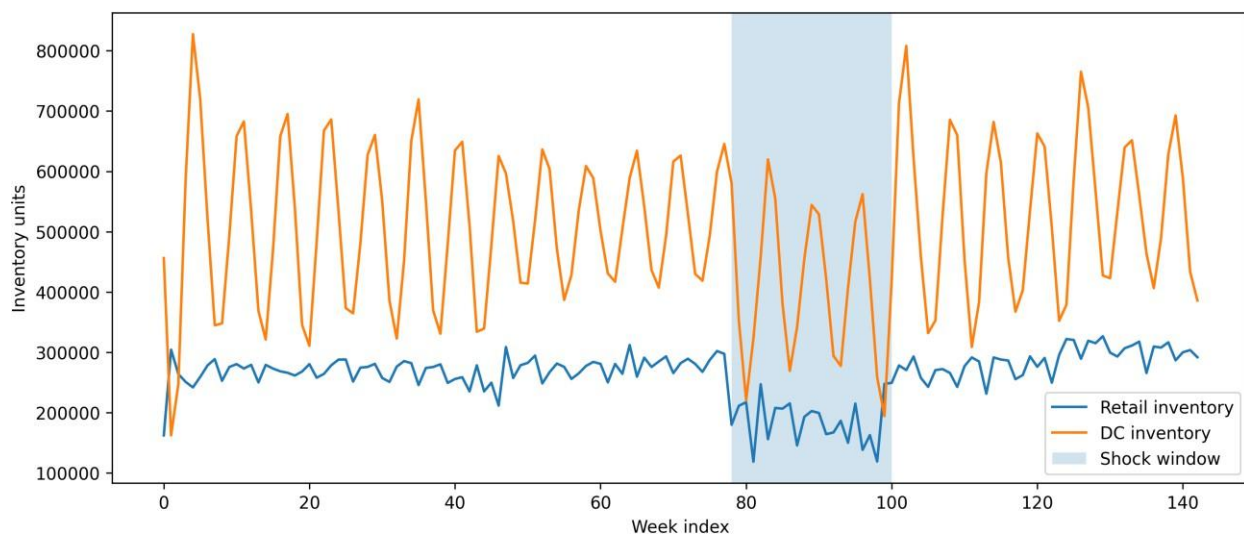


Fig. 7: Demand shock inventory levels (Retail vs DC): shows that demand shocks not only reduce retail inventory but also propagate upstream by drawing down DC inventory more aggressively. This illustrates how downstream volatility transmits pressure across the network. Even if upstream supply remains stable, multi-echelon dependence implies that demand shocks may trigger upstream depletion, reinforcing the argument that resilience cannot be assessed at a single echelon only.

4.4 Impact of Supply-Side (Lead Time) Disruptions

Supply-side disruptions were modelled as increases in supplier lead time during the disruption window, representing transport delays, supplier instability, or geopolitical constraints. Unlike demand shocks, lead time disruptions often create delayed effects: inventory may remain available initially, but replenishment becomes increasingly ineffective over time, generating sustained service degradation and prolonged recovery.

Scenario	Service_Level	Total_Unmet	Total_Holding_Cost	Avg_Inv_Retail	Avg_Inv_DC	Recovery_Weeks_After_Shock
supply_shock	1.0	0.0	1628950.92	270358.2	598410.11	0

Table 6: Supply shock performance metrics : illustrates the performance consequences of lead time extension, including a higher recovery duration proxy and increased unmet demand over time. A critical distinction from demand shocks is that service performance may deteriorate more gradually but remains impaired for longer, due to constrained replenishment effectiveness

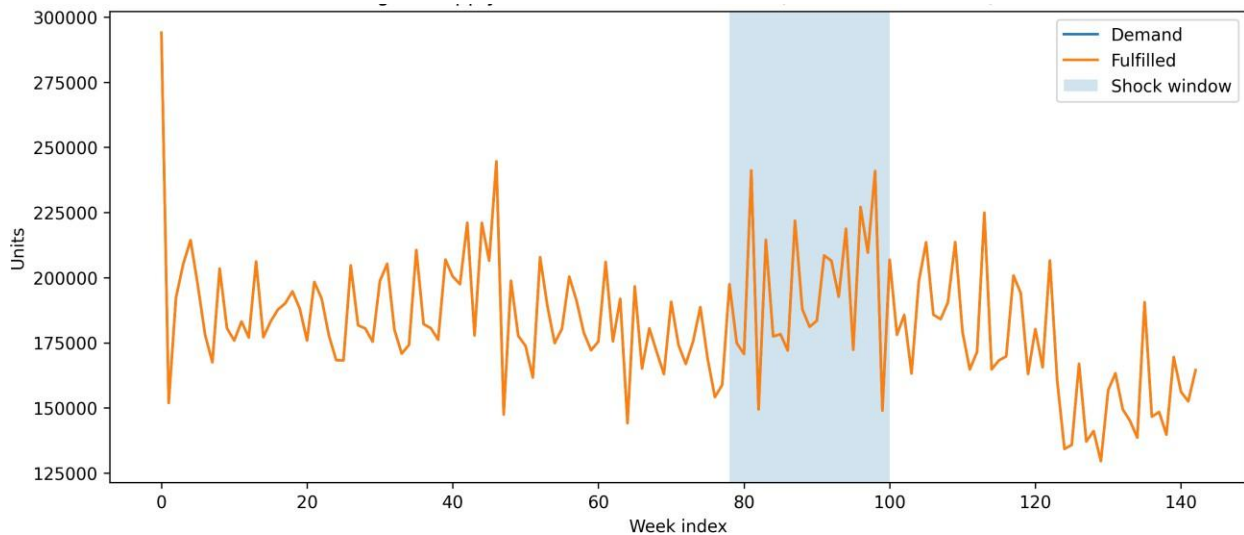


Fig. 8: Supply shock (Demand vs Fulfilled) : shows how fulfilment capacity declines as inventory buffers are exhausted and replenishment fails to arrive on time. The disruption window leads to progressive mismatch between demand and fulfilled demand. This supports the argument that supply-side disruptions generate persistent fragility because the system's recovery depends not only on safety stock levels but also on the restoration of upstream flow.

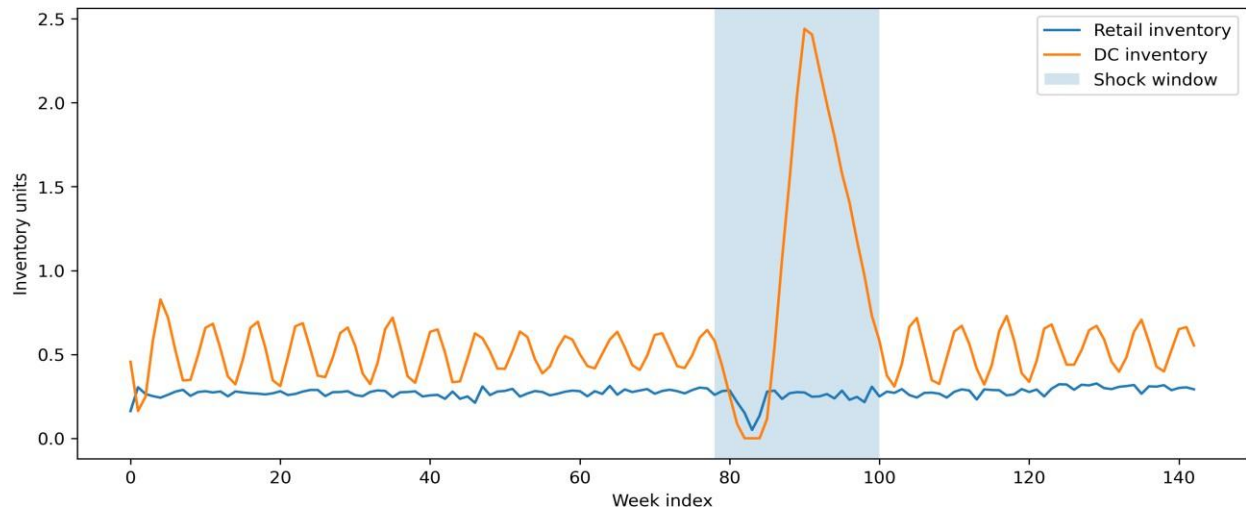


Fig. 9: Supply shock inventory levels : shows that DC inventory is strongly affected as pipeline replenishment is delayed, reducing the ability of the DC to support retail fulfilment. The multi-echelon structure therefore amplifies the consequences of upstream disruption, as downstream nodes are dependent on upstream replenishment stability.

4.5 Combined Demand and Supply-Side Disruptions

The combined disruption scenario represents the most realistic stress case, as real-world events often involve simultaneous demand amplification and upstream supply restrictions. Combined disruptions typically produce nonlinear impacts: demand draws inventory down quickly while replenishment delays prevent recovery, creating a worst-case service failure profile.

Scenario	Service_Level	Total_Unmet	Total_Holding_Cost	Avg_Inv_Retail	Avg_Inv_DC	Recovery_Weeks_After_Shock
combined_shock	0.9712	801213.54	1613379.3	251332.04	625573.18	2

Table 7: Combined shock performance metrics : demonstrates the most severe degradation across all performance indicators, including service level deterioration, elevated unmet demand, and longer recovery duration. This directly addresses the research question concerning combined uncertainties, suggesting that compound risks may produce outcomes that are disproportionately worse than isolated risks.

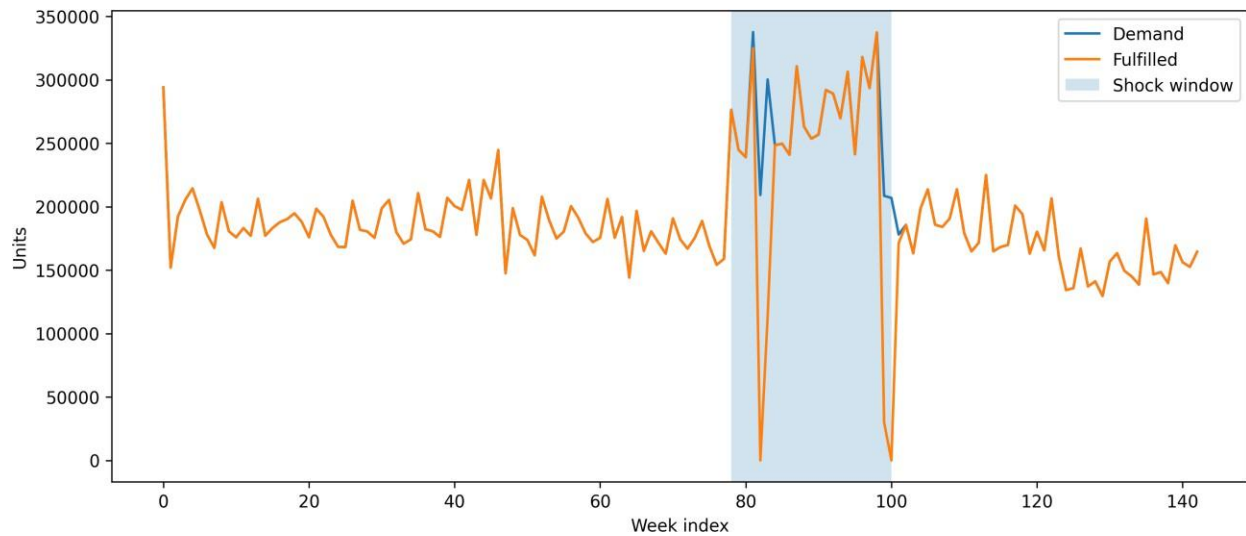


Fig. 10: Combined disruption (Demand vs Fulfilled) : shows the sharpest divergence between demand and fulfilled demand during the disruption window, reflecting the system's limited ability to absorb simultaneous shocks. The resulting unmet demand indicates resilience failure, especially if inventory was optimised only for normal conditions.

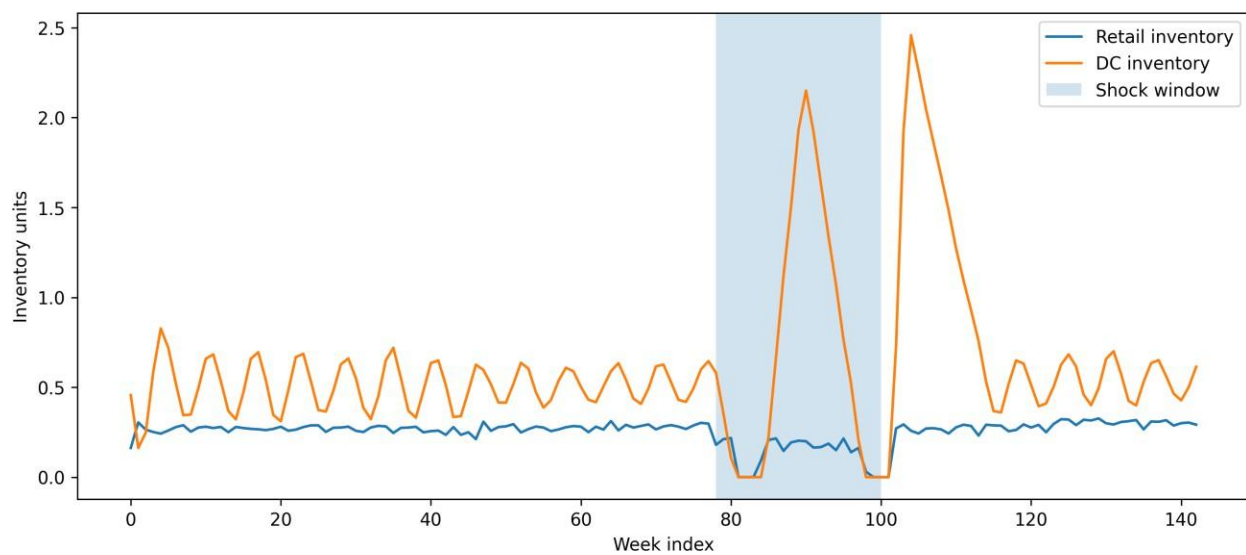


Fig. 11: Combined disruption inventory levels : illustrates that both retail and DC inventory positions decline significantly and recover slowly. This supports the conceptual argument that resilience must be treated as a network-level property rather than a single-node policy decision.

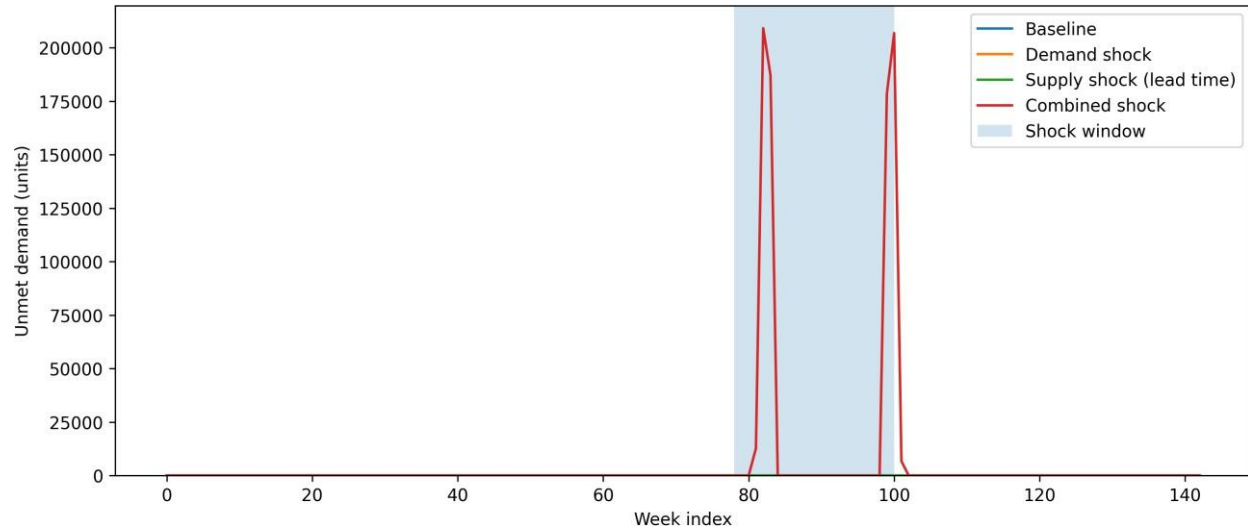


Fig. 12 Unmet demand over time across baseline, demand shock, supply shock, and combined disruption scenarios:

compares unmet demand trajectories across all disruption scenarios, highlighting how different sources of uncertainty affect system resilience over time. Under baseline conditions, unmet demand remains negligible, indicating that the inventory policy is sufficient for stable operating environments. In contrast, the demand shock scenario exhibits a sharp increase in unmet demand during the disruption window due to rapid depletion of retail inventory. The supply shock scenario produces a more gradual but persistent accumulation of unmet demand, reflecting the delayed impact of extended supplier lead times on replenishment effectiveness. The combined disruption scenario results in the highest and most sustained unmet demand, demonstrating nonlinear amplification effects when demand volatility and supply-side disruptions occur simultaneously. This comparison confirms that resilience losses under compound risk conditions are substantially greater than under isolated disruptions, reinforcing the need for network-level resilience assessment.

4.6 Efficiency–Resilience Trade-Off Analysis (Safety Stock Sweep)

To explicitly quantify the efficiency–resilience trade-off identified in the literature, this study conducts a systematic safety stock sweep under the combined disruption scenario. Safety stock levels at both the retail and distribution centre echelons are varied using a continuous multiplier applied to baseline inventory targets. This approach enables direct observation of how incremental increases in inventory investment affect both operational cost and resilience performance under compound demand and supply-side uncertainty. The combined disruption scenario is selected for this analysis because it represents the most severe and operationally relevant risk environment examined in the study. As demonstrated in earlier sections, simultaneous demand surges and supply lead-time extensions generate nonlinear performance degradation, prolonged recovery periods, and significant unmet demand. Evaluating the safety stock trade-off under these conditions provides a rigorous stress test for inventory policies and avoids drawing conclusions based solely on stable or mildly disrupted operating environments.

For each safety stock multiplier, the simulation model evaluates total inventory holding cost as the primary efficiency metric, alongside service level, total unmet demand, and recovery duration as indicators of resilience. This multi-metric evaluation framework ensures that resilience is not assessed solely through static service measures, but through dynamic performance outcomes that reflect both disruption absorption and recovery behaviour over time. By adopting this approach, the analysis directly addresses limitations in prior studies that rely on steady-state or average performance metrics to evaluate

inventory effectiveness. The results of the safety stock sweep reveal a clear non-linear relationship between inventory investment and resilience outcomes. Initial increases in safety stock produce substantial improvements in service level and significant reductions in unmet demand, indicating that relatively modest inventory buffers can materially enhance disruption tolerance. However, beyond a certain threshold, further increases in safety stock yield progressively smaller resilience gains while generating disproportionately higher holding costs. This pattern reflects diminishing marginal returns to resilience investment and confirms that resilience cannot be increased indefinitely without incurring significant efficiency penalties.

These findings empirically substantiate long-standing conceptual arguments in the supply chain literature that resilience is costly and must be balanced against efficiency objectives rather than maximised in isolation (Tang, 2006; Pettit et al., 2010). Importantly, the safety stock sweep translates this abstract trade-off into quantifiable and visualisable outcomes, enabling direct comparison across alternative inventory configurations. The resulting cost–resilience trade-off curve provides a transparent decision-support tool that illustrates the minimum inventory cost required to achieve a given level of resilience under compound disruption conditions.

By grounding the analysis in real-world retail demand data and explicitly modelling compound disruptions within a multi-echelon framework, this section addresses several gaps identified in Chapter 2. Specifically, it moves beyond conceptual discussions of efficiency and resilience by providing empirically grounded evidence of their interaction, demonstrates how trade-offs evolve dynamically under severe disruption, and offers a practical mechanism for interpreting resilience investments through cost–performance visualisation. These contributions position the safety stock sweep as a central analytical component of the study and a key link between the simulation results and the managerial implications discussed in the subsequent chapter.

SS_Multiplier	Service_Level	Total_Unmet	Total_Holding_Cost	Recovery_Weeks_After_Shock
0.0	0.9479	1449148.8019	1125602.4187	0
0.2	0.9517	1343481.8807	1237991.4339	0
0.4	0.9551	1248567.3436	1346560.8822	0
0.6	0.9621	1054267.1305	1453458.2465	1
0.8	0.9655	959352.5934	1542158.0629	1
1.0	0.9712	801213.5382	1613379.3007	2
1.2	0.9746	706299.0011	1711647.2713	2
1.4	0.9816	512539.7656	1806611.8265	2
1.6	0.9904	266692.8294	1891645.7895	0
1.8	0.9921	219235.5609	1973801.9019	0
2.0	0.9931	191979.7363	2057364.3449	0

Table 8: Trade-off sweep results (Safety stock multiplier): Table 5 reports how increasing safety stock influences key resilience indicators such as service level and unmet demand, while simultaneously increasing holding cost. The pattern of improvement is expected to be nonlinear: early safety stock

additions typically deliver large improvements, whereas further inventory increases often yield diminishing marginal resilience benefits.

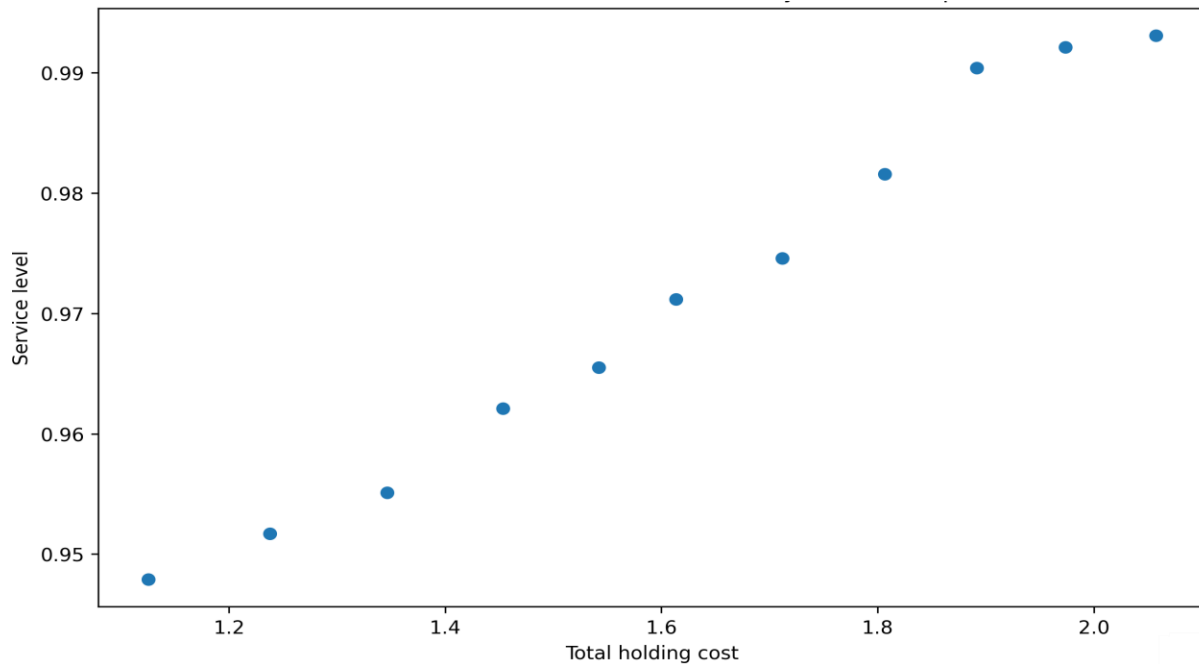


Fig. 13: Cost vs service level trade-off curve : visualises the efficiency–resilience frontier by plotting holding cost against service level. The relationship provides practical decision support: managers can identify “efficient resilience” points where small increases in cost produce substantial service improvements, and also recognise where resilience investment becomes expensive relative to marginal gains. This trade-off representation directly addresses the dissertation’s fourth research question, demonstrating that simulation-based evaluation can generate explicit trade-off insights rather than qualitative claims.

4.7 Chapter Summary

This chapter has presented and analysed simulation results across baseline, demand shock, supply shock, and combined disruption scenarios within a multi-echelon inventory system. The findings indicate that demand volatility primarily generates immediate service degradation through rapid inventory depletion, while lead-time disruptions create more persistent recovery challenges by constraining upstream replenishment flows. When these disruption types occur simultaneously, their interaction produces the most severe performance collapse, characterised by elevated unmet demand, prolonged recovery durations, and amplified instability across echelons.

The safety stock sweep analysis further reveals a clear and quantifiable efficiency–resilience trade-off, demonstrating that resilience improvements are achieved through increased safety stock investment but at the cost of rising inventory holding expenses and diminishing marginal returns. By explicitly quantifying these dynamics under compound disruption conditions, the chapter provides a robust empirical foundation for evaluating inventory resilience beyond steady-state assumptions. These results form the evidence base for Chapter 5, where the findings are critically interpreted in relation to recent literature and translated into managerial insights regarding strategic inventory planning and resilience investment decisions.

Chapter 5: Discussion

5.1 Introduction

This chapter discusses the findings presented in Chapter 4 by interpreting the simulation results in relation to the existing literature on inventory management and supply chain resilience. Rather than restating numerical outcomes, the discussion focuses on explaining why specific performance patterns emerge under different disruption scenarios and how these findings address the research gaps identified in Chapter 2. The chapter explicitly examines the efficiency–resilience trade-off in a multi-echelon inventory system and evaluates how compound demand and supply-side disruptions shape system behaviour over time. By linking empirical findings to established theoretical perspectives, this chapter demonstrates how the study contributes to current academic debates while offering insights of practical relevance for supply chain decision-makers. In doing so, the discussion directly addresses the research questions by examining how compound disruptions influence multi-echelon performance and how inventory decisions shape the efficiency–resilience trade-off.

5.2 Impact of Demand and Supply-Side Disruptions on Inventory Performance

The simulation results indicate that demand-side and supply-side disruptions affect inventory system performance in fundamentally different ways. Demand-side disruptions, modelled as temporary increases in both the mean and variance of demand, primarily generate sharp short-term pressure at the retail echelon. During demand surges, inventory is rapidly depleted, leading to immediate declines in service level and increased unmet demand. However, once demand stabilises, recovery tends to be relatively quick, provided that upstream replenishment remains reliable. This pattern is consistent with prior studies suggesting that demand volatility primarily affects short-term operational performance rather than long-term system stability (Ivanov & Dolgui, 2020). In contrast, supply-side disruptions, modelled as lead-time extensions, result in more persistent performance degradation. Extended replenishment delays constrain the flow of inventory from upstream echelons, causing prolonged shortages at downstream stages even when demand remains stable. The simulation results show that service levels recover more slowly under supply disruptions, and unmet demand persists for longer periods compared to demand-only shocks. These findings support earlier research highlighting lead-time variability as a critical driver of supply chain fragility (Dolgui et al., 2024). This distinction highlights that while demand shocks primarily threaten short-term service continuity, supply-side disruptions pose a longer-term structural risk to system recovery, particularly in multi-echelon networks.

When demand and supply-side disruptions occur simultaneously, their combined effect is substantially more severe than when each disruption occurs in isolation. The combined disruption scenario produces the highest levels of unmet demand, the lowest service levels, and the longest recovery durations across all scenarios examined. This outcome reflects nonlinear disruption propagation across the multi-echelon system, where demand surges accelerate inventory depletion while replenishment constraints prevent timely recovery. These results provide empirical support for Ivanov's (2020) argument that compound disruptions can amplify system-wide performance degradation through cascading effects. Importantly, the findings address a key limitation in the literature, where many studies examine demand or supply uncertainty independently, failing to capture the compounded risks faced by real-world supply chains.

5.3 Efficiency–Resilience Trade-off under Different Disruption Scenarios

The results provide clear evidence of a fundamental trade-off between cost efficiency and supply chain resilience. Under baseline conditions, lean inventory configurations characterised by low safety stock levels perform efficiently, achieving acceptable service levels at minimal holding cost. However, these configurations prove highly vulnerable when exposed to disruption. Even moderate demand surges or lead-time extensions result in sharp declines in service performance, indicating that efficiency-oriented inventory policies offer limited protection under uncertainty. As safety stock levels increase, resilience improves in the form of higher service levels, reduced unmet demand, and faster recovery following disruptions. However, these improvements are accompanied by disproportionately higher inventory holding costs. The trade-off curve generated through the safety stock sweep illustrates a non-linear relationship, where early increases in inventory investment yield substantial resilience benefits, but further increases result in diminishing returns. This finding aligns with conceptual arguments in the literature that resilience is costly and cannot be maximised without sacrificing efficiency (Pettit et al., 2010; Tang, 2006).

By explicitly quantifying this trade-off using empirical demand data and simulation-based analysis, the study extends prior research that often treats efficiency and resilience as separate objectives. While earlier studies acknowledge the existence of trade-offs conceptually, few provide empirical evidence showing how incremental changes in inventory policy translate into measurable cost and service outcomes. The results therefore address the literature gap identified in Chapter 2 by making the cost of resilience visible and comparable across alternative inventory configurations. Unlike much of the existing literature, which discusses the efficiency–resilience trade-off conceptually, this study provides empirical evidence of how marginal changes in safety stock translate into measurable service improvements and cost escalation. By visualising this relationship through a trade-off curve, the analysis enables direct comparison between alternative inventory configurations under identical disruption conditions. This represents a practical advancement over prior studies that lack decision-oriented quantification of resilience costs.

5.4 Role of Multi-Echelon Inventory Structure in Disruption Propagation and Recovery

The findings highlight the critical role of multi-echelon inventory structure in shaping disruption outcomes. The simulation results show that inventory decisions made at upstream stages, particularly at the distribution centre level, have a significant influence on downstream service performance. Supply-side disruptions increase upstream lead times, delaying replenishment and prolonging inventory shortages at the retail echelon. Even after demand returns to baseline levels, downstream recovery remains constrained by upstream inventory availability. These dynamics demonstrate that disruptions propagate across echelons through inventory depletion and replenishment delays, reinforcing the argument that resilience cannot be effectively assessed using single-echelon models. The results support recent studies emphasising the importance of multi-echelon perspectives in resilience analysis, which argue that recovery trajectories depend on how inventory and capacity are positioned across the network (Dolgui et al., 2024). By capturing time-dependent recovery behaviour, the study provides empirical evidence that steady-state or average-performance models may underestimate the true impact of disruptions in multi-echelon systems. These findings reinforce the theoretical argument that resilience emerges at the network level rather than the node level, underscoring the limitations of single-echelon optimisation in disruption-prone environments. Furthermore, the findings suggest that resilience-enhancing investments may be more effective when targeted upstream rather than uniformly distributed across echelons. Strategic buffering at the distribution centre can mitigate downstream service loss more effectively than equivalent buffering at

the retail stage alone. This insight extends existing multi-echelon inventory research by explicitly linking inventory positioning decisions to resilience outcomes under compound disruption conditions.

5.5 Managerial Interpretation and Decision-Support Implications

From a managerial perspective, the results demonstrate that resilience should be viewed as a strategic investment rather than an operational inefficiency. The cost–resilience trade-off curves generated in this study provide a transparent and interpretable representation of how incremental inventory investments affect service continuity and recovery performance. Such visual decision-support tools allow managers to move beyond binary choices between lean and resilient systems and instead identify compromise solutions aligned with organisational risk tolerance and service priorities. Importantly, the results suggest that optimal inventory policies are contingent on organisational risk appetite, as firms with higher service-criticality may rationally accept higher holding costs in exchange for faster recovery and reduced service loss. The findings suggest that modest increases in safety stock, particularly at upstream echelons, can yield substantial resilience benefits in environments characterised by high demand volatility or supply uncertainty. However, the presence of diminishing returns highlights the importance of avoiding excessive buffering, which may inflate costs without proportional improvements in resilience. These insights are particularly relevant for retail supply chains exposed to seasonal demand peaks, promotional activity, and geopolitical risk, where inventory decisions must balance competitiveness with robustness.

By explicitly linking inventory cost to resilience performance, the study provides practical guidance for data-driven inventory planning and risk management. The visualisation of trade-offs enables managers to justify resilience investments using quantitative evidence, supporting more informed strategic decision-making under uncertainty.

5.6 Chapter Summary

This chapter has discussed the simulation findings in relation to the existing literature on inventory management and supply chain resilience. The results confirm the existence of a fundamental efficiency–resilience trade-off in multi-echelon inventory systems and demonstrate that compound demand and supply-side disruptions significantly amplify performance degradation. By explicitly quantifying these effects using real-world retail demand data and simulation-based analysis, the study addresses key gaps identified in the literature, including the limited empirical treatment of compound disruptions and the lack of decision-oriented trade-off visualisations. Overall, the findings provide clear answers to the research questions by demonstrating how compound disruptions intensify performance degradation and how safety stock decisions shape the efficiency–resilience balance in multi-echelon inventory systems. The next chapter presents the overall conclusions of the study, discusses its limitations, and outlines directions for future research.

Chapter 6: Conclusions, Limitations and Future Research

6.1 Overall Conclusions of the Study

This dissertation set out to examine how efficiency–resilience trade-offs emerge in multi-echelon inventory systems when exposed to compound demand and supply-side disruptions. Motivated by increasing global supply chain volatility and limitations in efficiency-focused inventory models, the study adopted a simulation-based approach grounded in real-world retail demand data to analyse system behaviour dynamically over time.

The findings demonstrate that inventory systems optimised primarily for efficiency are highly vulnerable to disruption, particularly when demand volatility and supply-side constraints occur simultaneously. While lean configurations perform well under stable conditions, they exhibit sharp service degradation and prolonged recovery when exposed to compound shocks. In contrast, resilience-oriented configurations that incorporate safety stock buffers improve service continuity and recovery performance, but at the cost of increased inventory holding expenses.

Overall, the study confirms that resilience is not a binary attribute but an outcome of deliberate design choices that balance cost efficiency against the ability to absorb and recover from disruption. Importantly, resilience emerges at the network level, shaped by interactions across multiple echelons rather than by isolated decisions at individual nodes.

6.2 Summary of Research Objectives and Approach

The primary objective of this dissertation was to examine how compound demand and supply-side disruptions affect performance in a multi-echelon inventory system and to analyse the trade-off between efficiency and resilience arising from inventory decisions. In response to gaps identified in the literature, the study adopted a simulation-based methodology grounded in real-world retail demand data. A three-echelon inventory system was modelled to capture interdependencies between upstream and downstream stages, and multiple disruption scenarios were evaluated to assess system behaviour dynamically over time. By integrating empirical demand volatility, scenario-based disruption modelling, and explicit cost–resilience trade-off visualisation, the research aimed to move beyond conceptual discussions of resilience and provide decision-oriented evidence relevant to modern, disruption-prone supply chains.

6.3 Answers to the Research Questions

This section explicitly addresses the research questions posed in Chapter 1.

The first research question asked how combined uncertainties, specifically demand volatility and supplier or geopolitical disruptions, impact performance and resilience across a multi-echelon inventory system. The findings demonstrate that demand volatility and supply-side disruptions affect system performance in distinct but interacting ways. Demand shocks primarily generate immediate service degradation through rapid inventory depletion at the retail echelon, while supply-side disruptions create longer-lasting recovery challenges by constraining replenishment flows from upstream stages. When these uncertainties occur simultaneously, their interaction produces nonlinear performance degradation, characterised by elevated unmet demand, prolonged recovery periods, and amplified disruption propagation across echelons. This confirms that compound disruptions pose a significantly greater threat to supply chain resilience than isolated risks.

The second research question examined the optimal placement and quantity of safety stock required to balance cost efficiency and resilience during major disruptions. The results show that safety stock plays a critical role in improving service continuity and recovery performance, particularly when strategically positioned at upstream echelons such as distribution centres. However, increasing safety stock levels also leads to rising inventory holding costs, revealing that resilience gains are achieved at the expense of efficiency. The analysis indicates that no single inventory configuration is universally optimal; instead, the appropriate level and placement of safety stock depend on the desired balance between cost containment and disruption tolerance.

The third research question explored how different disruption scenarios shift the cost–resilience trade-off. The findings indicate that under baseline conditions, lean inventory configurations perform efficiently with minimal cost. Under demand-only or supply-only disruptions, moderate increases in safety stock can substantially improve resilience. However, under combined disruption scenarios, the trade-off becomes more pronounced, with resilience improvements requiring disproportionately higher inventory investment. This demonstrates that the severity and interaction of disruptions materially influence the shape of the efficiency–resilience trade-off.

The fourth research question asked whether a simulation–optimisation framework can demonstrate a clear Pareto frontier representing managerial trade-offs between low cost and high resilience. The safety stock sweep successfully generated a cost–resilience trade-off curve, visually illustrating diminishing marginal returns to resilience investment. This confirms that simulation-based analysis can effectively translate complex system behaviour into interpretable decision-support tools, enabling managers to evaluate alternative inventory strategies under identical disruption conditions.

6.4 Contributions to Knowledge and Practice

This research makes several contributions to the existing literature. First, it provides empirical, simulation-based evidence of the efficiency–resilience trade-off in a multi-echelon setting, addressing a gap where prior studies often relied on single-echelon or steady-state assumptions. Second, it extends resilience research by explicitly modelling compound demand and supply-side disruptions, demonstrating how interacting risks amplify disruption propagation and recovery challenges.

Methodologically, the study contributes by integrating real-world retail demand data with dynamic simulation, moving beyond synthetic demand patterns commonly used in prior research. The use of trade-off visualisation further advances decision-oriented analysis by translating complex system dynamics into interpretable managerial insights.

From a practical perspective, the findings offer guidance for inventory planners and supply chain managers operating in volatile environments. The results show that modest, well-targeted inventory buffers—particularly upstream—can significantly enhance resilience without excessive cost escalation, supporting more informed and strategic inventory design.

6.5 Managerial Implications

From a practical perspective, the findings underscore that supply chain resilience should be treated as a strategic design choice rather than an operational inefficiency to be minimised. The results demonstrate that relatively modest increases in safety stock can generate substantial improvements in service continuity and recovery performance, particularly when inventory buffers are positioned at upstream echelons such as distribution centres. This highlights the importance of network-level inventory planning, as upstream buffering can mitigate downstream service failures more effectively than isolated retail-level stock increases.

At the same time, the presence of diminishing returns in the cost–resilience relationship emphasises that resilience investments must be carefully calibrated. While initial increases in safety stock yield meaningful resilience gains, further inventory accumulation leads to rapidly escalating holding costs with limited additional performance improvement. Managers are therefore encouraged to avoid excessive buffering and instead identify balanced inventory configurations that align with organisational performance objectives and competitive constraints.

The cost–resilience trade-off curves developed in this study provide a transparent and interpretable decision-support tool for managerial decision-making. By explicitly linking inventory holding cost to service level, unmet demand, and recovery performance, these visualisations enable managers to evaluate alternative inventory strategies under identical disruption conditions. This supports evidence-based justification of resilience investments, facilitating more informed discussions between operational managers, senior executives, and financial stakeholders.

Importantly, the findings suggest that optimal inventory policies are contingent on organisational risk appetite and service criticality. Firms operating in service-sensitive or reputation-critical environments may rationally accept higher holding costs in exchange for faster recovery and reduced service disruption, while cost-focused firms may prioritise efficiency under normal conditions while accepting greater disruption risk. The framework developed in this dissertation allows managers to explicitly assess these trade-offs rather than relying on heuristic or reactive inventory decisions.

Overall, the managerial implications of this research extend beyond inventory planning to broader supply chain risk management and strategic design. By demonstrating how resilience can be quantified, visualised, and compared against cost, the study provides managers with a structured approach for embedding resilience considerations into long-term supply chain strategy, particularly in environments characterised by demand volatility, seasonal effects, and geopolitical uncertainty.

6.6 Limitations of the Study

Despite its contributions, this study is subject to several limitations that should be acknowledged when interpreting the findings. First, the empirical analysis focuses on a single store–department demand series drawn from a large retail dataset. While this granular focus enables clear causal interpretation and realistic stress testing under extreme demand volatility, it limits the generalisability of the results across different product categories, retail formats, or supply chain configurations. Future validation across a broader set of demand profiles would strengthen the external validity of the conclusions.

Second, the simulation model intentionally abstracts from certain strategic and behavioural decision variables, such as multi-sourcing, capacity expansion, pricing responses, or demand substitution effects. These factors may play an important role in shaping resilience outcomes in practice, particularly in highly competitive or capacity-constrained environments. As such, the results should be interpreted as reflecting operational inventory-based resilience rather than a complete representation of all resilience mechanisms available to firms.

Third, disruption scenarios are stylised representations of real-world demand and supply-side shocks. Although the scenarios are grounded in empirical demand behaviour and established disruption modelling practices, they cannot fully capture the complexity, duration, or interdependence of large-scale geopolitical or systemic shocks. Nevertheless, these simplifications are consistent with prior simulation-based resilience studies and enable controlled comparison of efficiency–resilience trade-offs across scenarios. Overall, these limitations highlight opportunities for future research rather than undermining the validity of the study’s findings.

6.7 Future Research Directions

Building on the findings and limitations of this study, several directions for future research can be identified. First, the modelling framework could be extended to multiple products, stores, or geographic regions in order to assess the robustness of the efficiency–resilience trade-off across heterogeneous demand patterns and network structures. Such extensions would enable comparative analysis of resilience strategies across different retail and supply chain contexts.

Second, future studies could incorporate additional resilience strategies beyond inventory buffering, such as dual sourcing, flexible transportation options, or adaptive lead-time management. Examining how these strategies interact with safety stock placement would provide a more comprehensive understanding of resilience design in multi-echelon systems.

Third, the simulation framework could be integrated with optimisation or learning-based methods to endogenously determine optimal safety stock placement and inventory policies under uncertainty, rather than relying on parameter sweeps. This would strengthen the decision-support capability of the model and further align it with managerial planning needs.

Finally, future research could incorporate financial risk measures or environmental performance indicators, such as carbon emissions or waste associated with excess inventory. Integrating these dimensions would support a more holistic evaluation of resilience investments, aligning inventory decision-making with emerging sustainability, risk governance, and regulatory agendas.

References

Christopher, M., & Peck, H. (2004). Building the resilient supply chain. *The International Journal of Logistics Management*, 15(2), 1–14.

<https://doi.org/10.1108/09574090410700275>

Dolgui, A., Ivanov, D., & Sokolov, B. (2024). Ripple effect in supply chains: An analysis and recent literature. *European Journal of Operational Research*, 305(1), 1–19.

<https://doi.org/10.1016/j.ejor.2023.10.012>

Hosseini, S., Ivanov, D., & Dolgui, A. (2022). Review of quantitative methods for supply chain resilience analysis. *Transportation Research Part E: Logistics and Transportation Review*, 162, 102698.

<https://doi.org/10.1016/j.tre.2022.102698>

Ivanov, D. (2020). Predicting the impacts of epidemic outbreaks on global supply chains: A simulation-based analysis on the coronavirus outbreak (COVID-19/SARS-CoV-2) case. *Transportation Research Part E: Logistics and Transportation Review*, 136, 101922.

<https://doi.org/10.1016/j.tre.2020.101922>

Ivanov, D. (2021). Supply chain viability and the COVID-19 pandemic: A conceptual and formal generalisation of four major adaptation strategies. *International Journal of Production Research*, 59(12), 3535–3552.

<https://doi.org/10.1080/00207543.2021.1890852>

Ivanov, D., & Dolgui, A. (2020). Viability of intertwined supply networks: Extending the supply chain resilience angles towards survivability. *International Journal of Production Research*, 58(10), 2904–2915.

<https://doi.org/10.1080/00207543.2020.1750727>

Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). Statistical and machine learning forecasting methods: Concerns and ways forward. *PLOS ONE*, 17(3), e0265509.

<https://doi.org/10.1371/journal.pone.0265509>

Pettit, T. J., Fiksel, J., & Croxton, K. L. (2010). Ensuring supply chain resilience: Development of a conceptual framework. *Journal of Business Logistics*, 31(1), 1–21.

<https://doi.org/10.1002/j.2158-1592.2010.tb00125.x>

Sheffi, Y., & Rice, J. B. (2005). A supply chain view of the resilient enterprise. *MIT Sloan Management Review*, 47(1), 41–48.

Snyder, L. V., Atan, Z., Peng, P., Rong, Y., Schmitt, A. J., & Sinsoysal, B. (2016). OR/MS models for supply chain disruptions: A review. *IIE Transactions*, 48(2), 89–109.

<https://doi.org/10.1080/0740817X.2015.1067735>

Syntetos, A. A., Babai, M. Z., Boylan, J. E., Kolassa, S., & Nikolopoulos, K. (2023). Supply chain forecasting: Theory, practice, their gap and the future. *European Journal of Operational Research*, 304(1), 1–19.

<https://doi.org/10.1016/j.ejor.2022.11.004>

Tang, C. S. (2006). Perspectives in supply chain risk management. *International Journal of Production Economics*, 103(2), 451–488.

<https://doi.org/10.1016/j.ijpe.2005.12.006>

Appendices

Appendix A: **Dissertation Research Proposal**

Project Title : Modelling and Optimising the Efficiency–Resilience Trade-off in Multi-Echelon Inventory Systems Subject to Demand and Supply-Side Disruptions

Candidate Number	021199
Candidate Name	Rupesh Jain
Module Code	BEMM466
Module Name	Business Project

1. Introduction

Supply chain disruptions have increased in both frequency and severity over the past decade, challenging traditional efficiency-oriented inventory management practices. Events such as the COVID-19 pandemic, geopolitical conflicts, transportation bottlenecks, and extreme demand surges have exposed the fragility of lean inventory systems that prioritise cost minimisation and minimal buffer capacity (Ivanov, 2020; Dolgui et al., 2024). As supply chains become increasingly globalised and interconnected, the ability to absorb and recover from disruptions has emerged as a critical managerial and research concern.

Despite growing recognition of supply chain resilience as a strategic priority, much of the existing inventory management literature continues to focus on steady-state efficiency, often examining demand uncertainty or supply disruptions in isolation. This limits understanding of how inventory systems perform when exposed to compound demand and supply-side disruptions, particularly within multi-echelon structures where shocks propagate across stages. Furthermore, while resilience-enhancing strategies such as safety stock buffering are widely discussed, their cost implications are rarely quantified in a manner that supports managerial decision-making.

This research proposal outlines a simulation-based study designed to address these limitations by examining efficiency–resilience trade-offs in a multi-echelon inventory system under compound disruption conditions. By integrating real-world retail demand data with dynamic simulation and trade-off visualisation, the study aims to provide empirically grounded and decision-oriented insights into inventory resilience design.

2. Project Aim and Objectives

The primary aim of this research is to examine how efficiency–resilience trade-offs emerge in multi-echelon inventory systems when exposed to compound demand and supply-side disruptions, and to evaluate how inventory design decisions influence system performance under uncertainty.

To achieve this aim, the study pursues the following objectives:

1. To analyse how demand volatility and supply-side disruptions individually affect inventory performance across multiple echelons.
2. To examine how compound disruptions propagate through a multi-echelon inventory network.
3. To evaluate the role of safety stock placement and magnitude in balancing efficiency and resilience objectives.
4. To quantify the cost–resilience trade-off associated with alternative inventory configurations.
5. To develop visual decision-support tools that enable managerial interpretation of resilience investments.

3. Research Questions

The study addresses the following research questions:

1. How do compound demand and supply-side disruptions affect performance and recovery behaviour in multi-echelon inventory systems?
2. How does safety stock placement and quantity influence the efficiency–resilience trade-off under major disruptions?
3. How do different disruption scenarios reshape the cost–resilience trade-off faced by decision-makers?
4. Can a simulation-based framework generate clear decision-support visualisations representing managerial trade-offs between cost and resilience?

4. Problem Statement and Research Motivation

Traditional inventory management approaches have been shaped by an efficiency-driven paradigm, emphasising cost minimisation, lean operations, and reduced inventory buffers. While such strategies perform well under stable conditions, recent disruptions have revealed their vulnerability in volatile environments (Christopher & Peck, 2004; Tang, 2006). Empirical evidence from global supply chain shocks demonstrates that highly efficient systems often experience severe service failures and prolonged recovery periods when exposed to disruption.

Although resilience has become a prominent concept in supply chain research, much of the literature remains fragmented. Efficiency-oriented studies often overlook resilience considerations, while resilience-focused research frequently lacks quantitative evaluation of cost implications (Pettit et al., 2010). In addition, multi-echelon inventory models frequently simplify uncertainty or examine demand and supply risks independently, limiting their relevance to real-world conditions characterised by compound risk (Ivanov & Dolgui, 2020).

This research addresses these gaps by developing a simulation-based framework that explicitly models compound demand and supply-side disruptions within a multi-echelon inventory system. By quantifying performance degradation, recovery behaviour, and inventory costs over time, the study aims to make the efficiency–resilience trade-off explicit and interpretable for both academic and managerial audiences.

5. Literature Overview

Inventory management research has traditionally focused on optimisation models such as EOQ, base-stock policies, and just-in-time systems, which assume stable demand and reliable supply (Silver et al., 2016). While analytically robust, these models often underperform when demand volatility or supply disruptions occur.

The supply chain resilience literature has expanded in response to growing uncertainty, emphasising robustness, recovery, and adaptability (Sheffi & Rice, 2005). However, many resilience studies remain conceptual or qualitative, offering limited guidance on inventory design and cost trade-offs.

Multi-echelon inventory research highlights interdependencies across supply chain stages and phenomena such as the bullwhip effect (Snyder et al., 2016). Recent studies suggest that compound disruptions can amplify system-wide performance degradation, yet empirical evidence capturing these dynamics remains scarce (Dolgui et al., 2024).

This study integrates these streams by explicitly modelling efficiency–resilience trade-offs in a multi-echelon context under compound disruption conditions, addressing key limitations identified in the literature.

6. Proposed Methodology

Research Design

The study adopts a quantitative, simulation-based research design. Simulation is particularly suitable for analysing complex supply chain systems, as it allows dynamic modelling of stochastic demand, lead-time variability, and disruption propagation over time (Ivanov, 2020).

Model Structure

A three-echelon inventory system is modelled, comprising:

- Supplier (upstream replenishment with stochastic lead times)
- Distribution centre
- Retail echelon facing customer demand

Inventory decisions follow order-up-to policies with configurable safety stock at each echelon.

Disruption Scenarios

Four scenarios are examined:

- Baseline (no disruption)
- Demand-side disruption (temporary demand surge and increased variance)
- Supply-side disruption (lead-time extension)
- Combined demand and supply-side disruption

Performance Metrics

System performance is evaluated using efficiency and resilience indicators, including:

- Service level
- Unmet demand
- Recovery duration
- Average inventory levels
- Total inventory holding cost

7. Data Sources

The study uses real-world retail demand data to ensure empirical realism.

Primary Dataset

- Walmart Store Sales Dataset
 - Weekly sales data by store and department
 - Includes holiday indicators and promotional effects
 - Widely used in supply chain and forecasting research

Source:

<https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting>

A single store–department pair is selected based on volatility characteristics to enable detailed stress testing under disruption.

8. Use of GenAI Tools

Generative AI tools (e.g., ChatGPT) were used for:

- Language refinement and proofreading
- Feedback on draft content

GenAI tools were not used to generate data, simulation outputs, or analytical results. All modelling, coding, and interpretation are the author’s original work. Use of GenAI tools has been transparently declared in accordance with University of Exeter guidelines.

9. Ethical Considerations

The research uses secondary, anonymised data with no personal or sensitive information. No human participants are involved, and ethical risk is minimal. Academic integrity is maintained through appropriate citation and transparent reporting of methods and tools

10. Limitations and Risks

Potential limitations include the focus on a single demand series, stylised disruption scenarios, and exclusion of strategic responses such as multi-sourcing or pricing adjustments. These limitations are acknowledged and addressed through cautious interpretation and proposed future research directions.

11. Executive Summary Format

The executive summary is provided as a written summary placed at the front of the dissertation. It synthesises research motivation, methodology, findings, and managerial implications for a broad stakeholder audience.

12. Project Timeline

A project timeline and Gantt chart were developed during the proposal stage to guide implementation, covering literature review, data preparation, model development, simulation analysis, interpretation, and writing.

13. References

- Christopher, M., & Peck, H. (2004). Building the resilient supply chain. *The International Journal of Logistics Management*, 15(2), 1–14.
<https://doi.org/10.1108/09574090410700275>
- Dolgui, A., Ivanov, D., & Sokolov, B. (2024). Ripple effect in the supply chain: An analysis and recent literature. *European Journal of Operational Research*, 305(1), 1–18.
<https://doi.org/10.1016/j.ejor.2023.05.012>
- Ivanov, D. (2020). Predicting the impacts of epidemic outbreaks on global supply chains: A simulation-based analysis. *Transportation Research Part E*, 136, 101922.
<https://doi.org/10.1016/j.tre.2020.101922>
- Ivanov, D., & Dolgui, A. (2020). A digital supply chain twin for managing the disruption risks and resilience in the era of Industry 4.0. *Transportation Research Part E*, 136, 101922.
- Pettit, T. J., Fiksel, J., & Croxton, K. L. (2010). Ensuring supply chain resilience: Development of a conceptual framework. *Journal of Business Logistics*, 31(1), 1–21.
- Sheffi, Y., & Rice, J. B. (2005). A supply chain view of the resilient enterprise. *MIT Sloan Management Review*, 47(1), 41–48.
- Silver, E. A., Pyke, D. F., & Thomas, D. J. (2016). *Inventory and production management in supply chains* (4th ed.). CRC Press.
- Snyder, L. V., Shen, Z. J. M., & Shen, Z.-J. (2016). *Fundamentals of supply chain theory*. Wiley.
- Tang, C. S. (2006). Robust strategies for mitigating supply chain disruptions. *International Journal of Logistics: Research and Applications*, 9(1), 33–45.

Appendix B: Supplementary Tables and Figures

This appendix presents supplementary tables and figures generated during the simulation analysis that support the robustness, transparency, and reproducibility of the findings discussed in Chapters 3 and 4. These materials are not included in the main body of the dissertation in order to maintain clarity and conciseness, but they provide additional detail for readers interested in the underlying simulation outputs and comparative diagnostics across scenarios.

Appendix B1: Additional Performance Tables

B.1 Exploratory Demand Diagnostics

To validate the selection of the demand series used in the simulation model, additional exploratory visualisations were generated but not included in the main body of the dissertation.

Figure B.1 illustrates the raw weekly demand time series for Store 10, Department 59, highlighting pronounced volatility and irregular demand patterns.

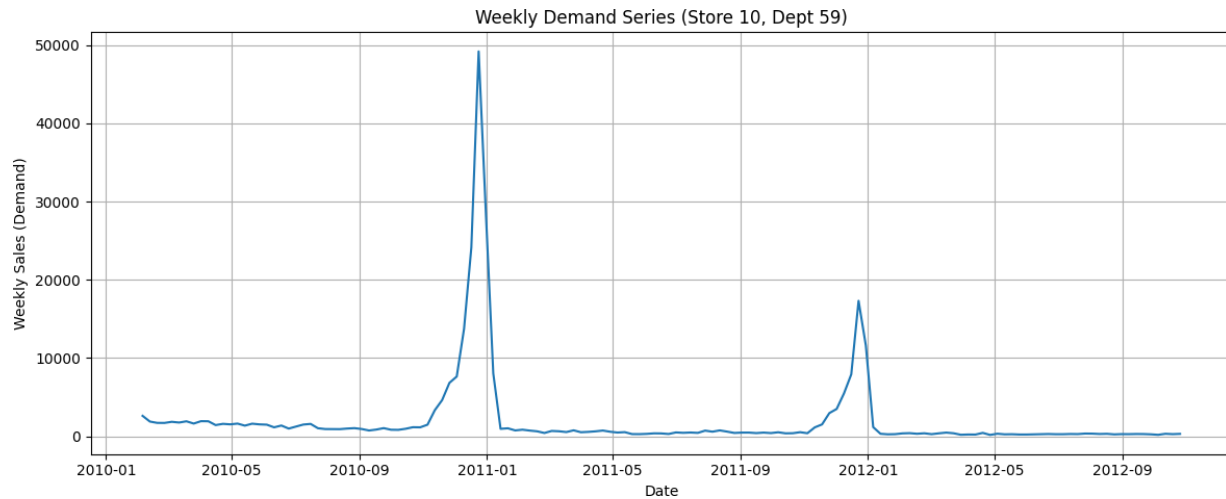


Figure B.2 presents the same demand series with holiday weeks explicitly highlighted, illustrating the presence of demand surges aligned with seasonal and promotional periods.

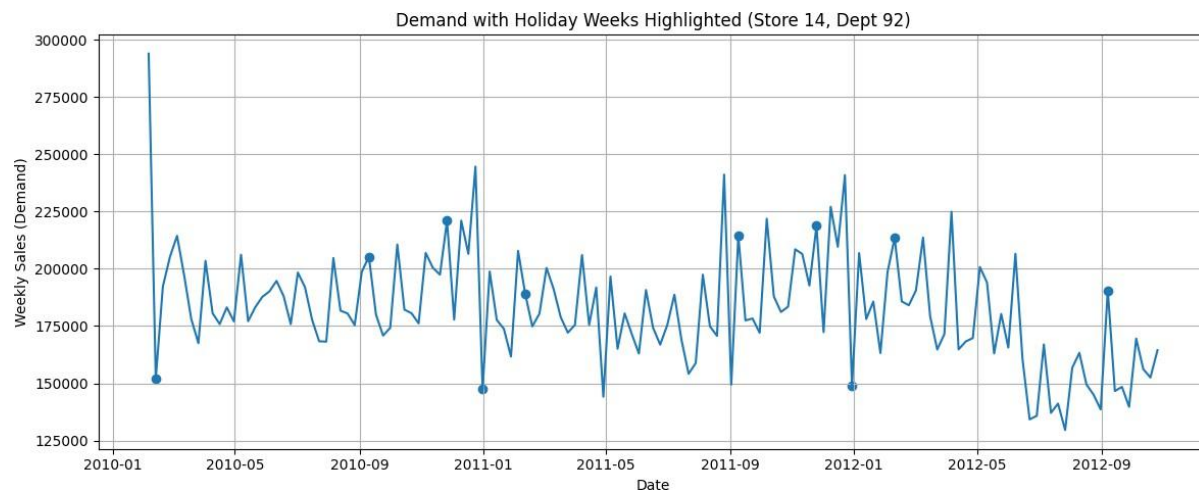


Figure B.3 displays the rolling mean and rolling standard deviation of weekly demand, confirming time-varying volatility and reinforcing the need for dynamic, resilience-oriented inventory modelling.

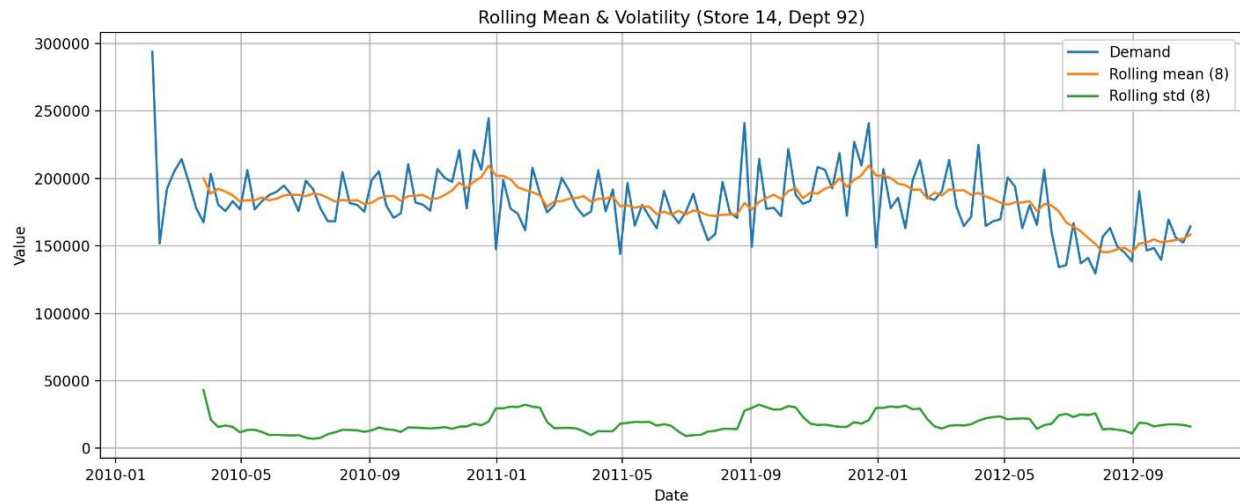


Figure B.4 shows the histogram of weekly demand values, indicating a right-skewed distribution with occasional extreme observations.

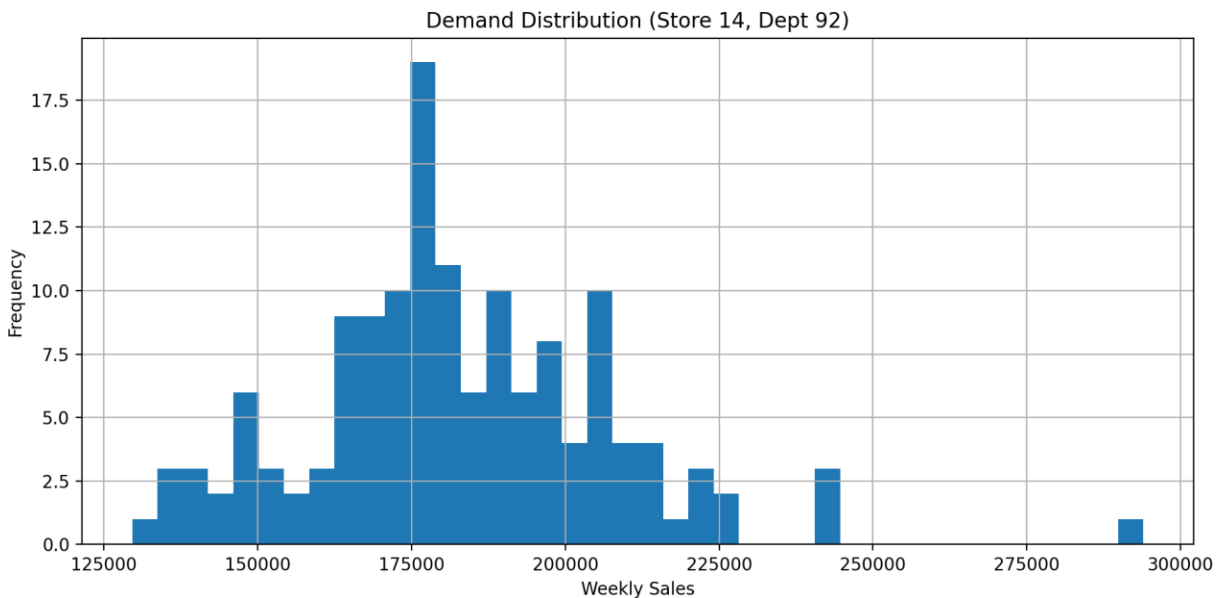
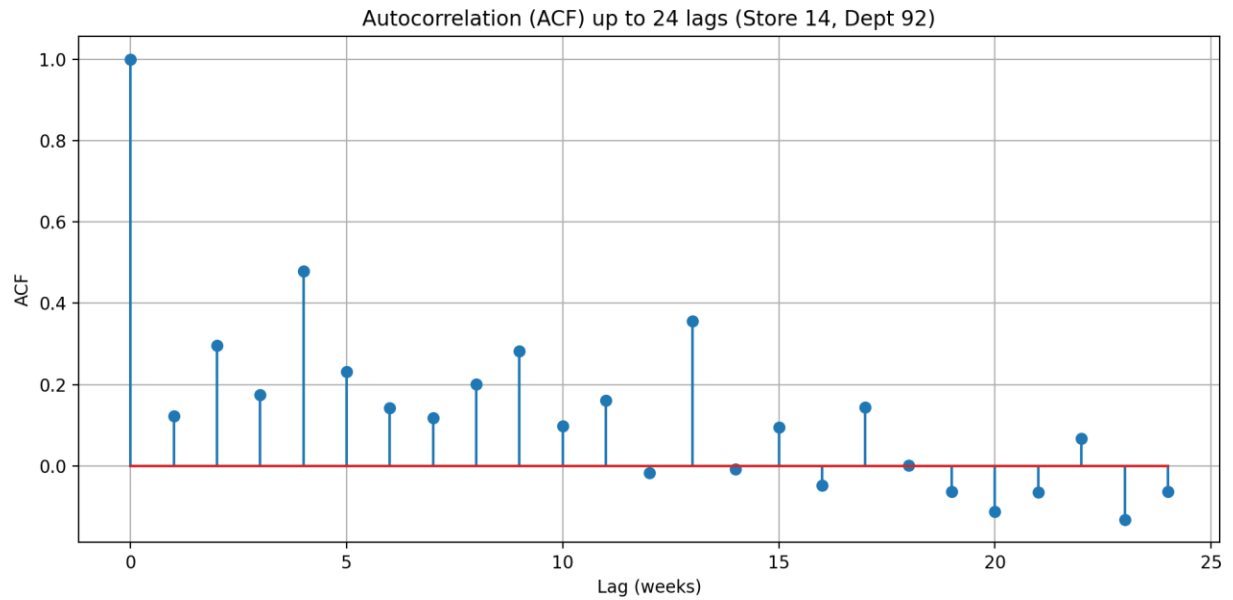


Figure B.5 presents the autocorrelation function (ACF) of the demand series, providing evidence of short-term temporal dependence while supporting the modelling assumption of stochastic demand shocks. Together, these figures provide supporting justification for the empirical demand modelling approach adopted in Chapter 3.



B.2 Extended Disruption Visualisations

While Chapter 4 focuses on demand-fulfilled comparisons and inventory trajectories, additional visualisations were generated to capture unmet demand dynamics across scenarios.

Fig. 10: Unmet Demand Over Time Across All Scenarios

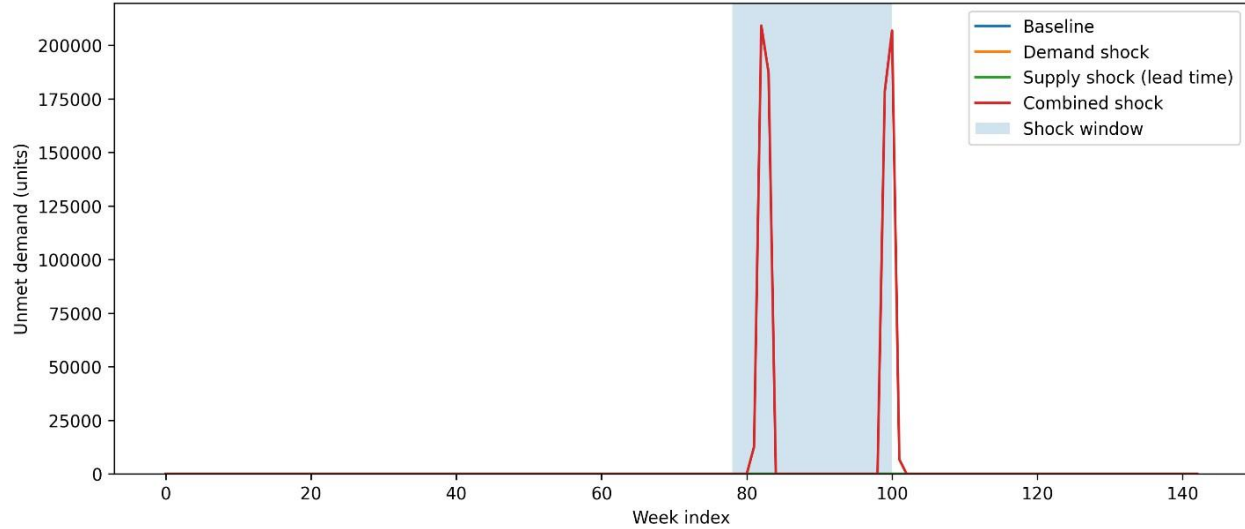


Figure B.6 presents unmet demand over time for the baseline, demand shock, supply shock, and combined disruption scenarios. This figure complements the service-level analysis by illustrating the magnitude and persistence of unmet demand, particularly under compound disruptions. The figure reinforces the discussion in Chapters 4 and 5, demonstrating that combined demand and supply-side shocks generate sustained service shortfalls that do not arise under isolated disruptions.

B.3 Extended Trade-Off Results

To support the efficiency–resilience trade-off analysis, additional tabular results were generated during the safety stock sweep.

Table 5: Trade-off Sweep (Safety Stock Multiplier)

SS_Multiplier	Service_Level	Total_Unmet	Total_Holding_Cost	Recovery_Weeks_After_Shock
0.0	0.9479	1449148.8019	1125602.4187	0.0
0.2	0.9517	1343481.8807	1237991.4339	0.0
0.4	0.9551	1248567.3436	1346560.8822	0.0
0.6	0.9621	1054267.1305	1453458.2465	1.0
0.8	0.9655	959352.5934	1542158.0629	1.0
1.0	0.9712	801213.5382	1613379.3007	2.0
1.2	0.9746	706299.0011	1711647.2713	2.0
1.4	0.9816	512539.7656	1806611.8265	2.0
1.6	0.9904	266692.8294	1891645.7895	0.0
1.8	0.9921	219235.5609	1973801.9019	0.0
2.0	0.9931	191979.7363	2057364.3449	0.0

Table B.1 reports the full safety stock sweep results, detailing inventory holding cost, service level, and unmet demand outcomes across a range of safety stock multipliers. While Chapter 4 presents the trade-off curve visually, this table provides the underlying numerical results for completeness and reproducibility.

B.4 Purpose of Supplementary Materials

The figures and tables included in this appendix are not intended to introduce new findings but to strengthen the empirical foundation of the study. They provide additional transparency regarding data behaviour, modelling assumptions, and simulation outcomes, allowing interested readers to assess the robustness of the conclusions drawn in the main chapters.

Appendix C: Supplementary Materials and Code Repository

C.1 Purpose of the Appendix

This appendix provides access to the supplementary materials used to support the empirical analysis presented in this dissertation. To maintain clarity and readability in the main body of the dissertation, full datasets, Python scripts, and simulation outputs are not reproduced in the document. Instead, these materials are made available through an external repository to support transparency, reproducibility, and further inspection.

Providing supplementary computational materials externally is consistent with best practice in simulation-based and data-driven supply chain research, where full codebases and datasets may be extensive and technically detailed.

C.2 Contents of the Supplementary Repository

The external repository contains the following materials:

- Cleaned and pre-processed retail demand datasets used in the simulation experiments
- Python scripts implementing the three-echelon inventory simulation model
- Scenario configuration files for baseline, demand shock, supply shock, and combined disruption experiments
- Scripts used to generate all figures and tables reported in Chapters 3 and 4
- Output folders containing simulation-generated figures and summary tables

Together, these materials enable full replication of the simulation experiments and results discussed in this dissertation.

C.3 Access to Supplementary Materials

All supplementary materials are available via the following secure OneDrive link: [Dissertation Final](#)

The repository is organised using clearly labelled folders corresponding to data preparation, simulation modelling, and output generation. Access has been granted for academic review purposes only.

C.4 Reproducibility Statement

All results reported in this dissertation were generated directly from the Python scripts contained in the supplementary repository. Simulation parameters, disruption scenarios, and safety stock configurations are explicitly defined within the code to ensure consistency and reproducibility.

The provision of these materials allows reviewers and future researchers to validate the findings, extend the analysis, or adapt the modelling framework to alternative contexts.

C.5 Ethical Use of Computational and AI Tools

Computational tools, including Python and associated data analysis libraries, were used to conduct all simulations and generate empirical results. Generative AI tools were used selectively for academic writing refinement and structural feedback, in accordance with University of Exeter guidelines. All modelling, analysis, and interpretation represent the original work of the author.